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INTEGRATED CIRCUIT DEVICE AND NON-TRANSITORY COMPUTER-READABLE RECORDING MEDIUM STORING CELL LAYOUT FOR INTEGRATED CIRCUIT DEVICE

Abstract

An integrated circuit (IC) device includes a substrate, at least one active region over the substrate, at least one gate region extending across the at least one active region, and a metal layer. The at least one active region and the at least one gate region configure a plurality of transistors electrically coupled into a circuit having an input and an output. The metal layer includes a first conductive pattern electrically coupled to the input of the circuit, and a second conductive pattern electrically coupled to the output of the circuit. The first conductive pattern and the second conductive pattern extend obliquely to the at least one active region or the at least one gate region.

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Background/Summary

RELATED APPLICATION(S) [0001] The instant application is a continuation of U.S. application Ser. No. 18/355,273, filed Jul. 19, 2023, which is a continuation of U.S. application Ser. No. 17/313,474, filed May 6, 2021, now U.S. Pat. No. 11,715,733, issued Aug. 1, 2023. The above-referenced application(s) and patent(s) are incorporated by reference herein in their entireties.

BACKGROUND

[0002] An integrated circuit (“IC”) device includes one or more semiconductor devices represented in an IC layout diagram (also referred to as “layout diagram”). A layout diagram is hierarchical and includes modules which carry out higher-level functions in accordance with the semiconductor device's design specifications. The modules are often built from a combination of cells, each of which represents one or more semiconductor structures configured to perform a specific function. Cells having pre-designed layout diagrams, sometimes known as standard cells, are stored in standard cell libraries (hereinafter “libraries” or “cell libraries” for simplicity) and accessible by various tools, such as electronic design automation (EDA) tools, to generate, optimize and verify designs for ICs.

[0003] Layout diagrams are generated in a context of design rules. A set of design rules imposes constraints on the placement of corresponding patterns in a layout diagram, e.g., geographic/spatial restrictions, connectivity restrictions, or the like. Often, a set of design rules includes a subset of design rules pertaining to the spacing and other interactions between patterns in adjacent or abutting cells where the patterns represent conductors in a layer of metallization. Routing and placing is where the different semiconductor devices in an IC device are connected. One of the goals of routing and placing in a layout is to reduce the amount of routing required and thereby improve power and space consumed by a semiconductor device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is a circuit diagram of a cell, in accordance with some embodiments.

[0006] FIGS. 2A-2C are schematic views at various layers of a layout diagram of a cell, and FIG. 2D is a schematic view of the layout diagram of the cell including the layers in FIGS. 2A-2C, in

accordance with some embodiments.

[0007] FIG. 3 is a schematic cross-sectional view, taken along lines III-III in FIGS. 2A-2D, of an IC device, in accordance with some embodiments.

[0008] FIG. 4 is a circuit diagram of a cell, in accordance with some embodiments.

[0009] FIGS. 5A-5C are schematic views of layout diagrams of various cell, in accordance with some embodiments.

[0010] FIGS. 6A-6C are schematic views of various layout diagrams of a cell, in accordance with some embodiments.

[0011] FIG. 6D is a schematic view showing cell boundaries of a cell, in accordance with some embodiments.

[0012] FIG. 7 is a circuit diagram of a cell, in accordance with some embodiments.

[0013] FIGS. 8A-8B are schematic views of various layout diagrams of a cell, in accordance with some embodiments.

[0014] FIG. 9 is a schematic cross-sectional view, taken along line IX-IX in FIG. 8B, of an IC device, in accordance with some embodiments.

[0015] FIGS. 10A-10E are each a schematic cross-sectional view of an IC layout diagram of an IC device including layout diagrams of various cells constituting the IC layout diagram of the IC device, in accordance with some embodiments.

[0016] FIGS. 11A-11I are each a schematic cross-sectional view of a layout diagram of a filler cell, in accordance with some embodiments.

[0017] FIG. 12 is a schematic cross-sectional view of an IC layout diagram of an IC device including layout diagrams of various cells constituting the IC layout diagram of the IC device, in accordance with some embodiments.

[0018] FIGS. 13A-13C are flowcharts of various methods, in accordance with some embodiments.

[0019] FIG. 14 is a block diagram of an electronic design automation (EDA) system in accordance with some embodiments.

[0020] FIG. 15 is a block diagram of an IC device manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

[0021] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, materials, values, steps, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Source/drain(s) may refer to a source or a drain, individually or collectively dependent upon the context.

[0022] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0023] A cell has one or more active regions and gate regions electrically coupled to form internal circuitry of the cell. At an upper metal layer (also referred to as “pin access layer”), the cell includes one or more input/output (IO) pins (also referred to as “IO patterns”) configured to electrically couple the internal circuitry inside the cell to external circuitry outside the cell. The cell has a boundary in which the one or more IO patterns are arranged. In some embodiments, at least one side or edge of the boundary and at least one IO pattern of the cell are slanted or oblique to the gate regions and/or the active regions. As a result, in at least one embodiment, it becomes easier to access, or electrically couple, to the oblique IO pattern from a further metal layer over the pin access layer. This is an improvement over other approaches in which cell boundaries are rectangular and IO patterns are oriented parallel to active regions or gate regions.

[0024] FIG. 1 is a circuit diagram of a cell **100**, in accordance with some embodiments. In FIG. 1, the cell **100** is an inverter. This is an example, and other cells are within the scope of various embodiments. For example, in various embodiments, the cell **100** is a function cell, an engineering change order (ECO) cell, a physical cell, a filler cell, or another type of cell or combination of cells capable of being defined in an IC layout diagram.

[0025] A function cell is a cell pre-designed to provide a specific function to an IC incorporating such a function cell. Examples of function cells include, but are not limited to, a logic gate cell, a memory cell, or the like. Examples of logic gate cells include, but are not limited to, AND, OR, NAND, NOR, XOR, INV, AND-OR-Invert (AOI), OR-AND-Invert (OAI), MUX, Flip-flop, BUFF, Latch, delay, clock, or the like. Examples of memory cells include, but are not limited to, a static random access memory (SRAM), a dynamic RAM (DRAM), a resistive RAM (RRAM), a magnetoresistive RAM (MRAM), a read only memory (ROM) cell, or another type of cell capable of having multiple states representative of logical values.

[0026] An ECO cell is a cell pre-designed without a specific function, but is programmable to provide an intended function. For example, to design an IC, the pre-designed layouts of one or more function cells are read out from a standard cell library and placed into an initial IC layout. The IC layout also includes one or more ECO cells which are not yet connected or routed to the function cells. When the IC layout is to be revised, one or more of the already placed ECO cells are programed to provide an intended function and routed to the function cells. The programing of ECO cells involves modifications in one or more layers of the IC layout and/or masks for manufacturing the IC.

[0027] A physical cell is a cell configured to provide a function, other than a logic function, to an IC incorporating such physical cell. Examples of physical cells include, but are not limited to, a TAP cell, a DCAP cell, or the like. A TAP cell defines a region in a doped well where the doped well is coupled to a bias voltage, such as a power supply voltage. TAP cells are included in an IC layout diagram, e.g., to improve latch-up immunity of ICs manufactured in accordance with the IC layout diagram. A DCAP cell includes one or more decoupling capacitors (decap) between power buses or rails, e.g., as a charge reservoir to provide additional power in situations where there is a high demand for current from the power supply.

[0028] A purpose of filler cells is to fill an empty space in an IC layout diagram, for example, to satisfy one or more design rules, such as a minimum spacing between adjacent features. In some embodiments, a physical cell is placed as a filler cell. In at least one embodiment, a filler cell is a cell with no logical functionality that is not connected or routed to other cells in an IC layout diagram. Cells other than filler cells are referred to herein as “non-filler cells.”

[0029] In the example circuit diagram in FIG. 1, the inverter in the cell **100** comprises a p-channel metal-oxide semiconductor (PMOS) transistor PM and an n-channel metal-oxide semiconductor (NMOS) transistor NM coupled in series between a first power supply voltage VDD and a second power supply voltage VSS. Specifically, the transistor PM comprises a gate region GP, a source region SP, and a drain region DP. The transistor NM comprises a gate region GN, a source region SN, and a drain region DN. The gate regions GP, GN are coupled to an input node IN, also referred

to herein or indicated in the drawings as “I.” The drain regions DP, DN are coupled to an output node OUT, also referred to herein or indicated in the drawings as “ZN.” The source region SP is coupled to VDD, and the source region SN is coupled to VSS. In at least one embodiment, VDD is a positive power supply voltage, and VSS is a ground voltage. The inverter is configured to invert a signal at the input node IN and to output the inverted signal at the output node OUT. Other types of transistors are within the scopes of various embodiments, as described herein.

[0030] FIGS. 2A-2C are schematic views at various layers of a layout diagram of a cell **200**, in accordance with some embodiments. FIG. 2D is a schematic view of the layout diagram of the cell **200** including the layers in FIGS. 2A-2C. In at least one embodiment, the layout diagram of the cell **200** in FIG. 2D is stored as a standard cell in a standard cell library on a non-transitory computer-readable recording medium. The cell **200** is an inverter corresponding to the cell **100**. In the example configuration in FIGS. 2A-2D, the cell **200** is an inverter with a driving strength of 1, also referred to as “INVD1.” Other example inverter cells with different driving strengths are herein, e.g., described with respect to FIGS. 6A-6C.

[0031] FIG. 2B is a schematic view showing several layers of the cell **200**. In FIG. 2A, the cell **200** comprises at least one active region, at least one gate region extending across the at least one active region, and a first boundary in which the at least one active region and the at least one gate region are arranged. For example, the cell **200** comprises active regions **201**, **202**, a gate region **110**, and a first boundary **220**.

[0032] The active regions **201**, **202** are arranged inside the first boundary **220**, and extend along a first axis, i.e., X-X' axis. Active regions are sometimes referred to as oxide-definition (OD) regions, and are schematically illustrated in the drawings with the label “OD.” In an IC device comprising the cell **200** in accordance with at least one embodiment, the active regions **201**, **202** are over a first side, or a front side, of a substrate as described herein. The active regions **201**, **202** include P-type dopants and/or N-type dopants to form one or more circuit elements or devices. Examples of circuit elements include, but are not limited to, transistors and diodes. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, P-channel and/or N-channel field effect transistors (PFETs/NFETs), etc.), FinFETs, planar MOS transistors with raised source/drains, nanosheet FETs, nanowire FETs, or the like. An active region configured to form one or more PMOS devices therein is referred to herein as “PMOS active region,” and is schematically indicated in the drawings with the label “PMOS.” An active region configured to form one or more NMOS devices therein is referred to herein as “NMOS active region,” and is schematically indicated in the drawings with the label “NMOS.” In embodiments described herein, a PMOS active region is replaceable with an NMOS active region, and vice versa.

[0033] The gate region **210** is arranged inside the first boundary **220**, and extends across the active regions **201**, **202** along a second axis, i.e., Y-Y' axis, which is transverse to the X-X' axis. In at least one embodiment, the Y-Y' axis is perpendicular to the X-X' axis. The gate region **210** includes a conductive material, such as, polysilicon, and is schematically illustrated in the drawings with the label “PO.” Other conductive materials for the gate region, such as metals, are within the scope of various embodiments.

[0034] In the example configuration in FIG. 2A, the active region **201** is a PMOS active region inside an N well **207** and configured to form, together with the gate region **210**, the transistor PM of the inverter as described with respect to FIG. 1. The active region **202** is an NMOS active region configured to form, together with the gate region **210**, the transistor NM of the inverter as described with respect to FIG. 1. Specifically, the active region **201** comprises a source region **203** and a drain region **205** on opposite sides of a first section of the gate region **210** which extends over the active region **201**. The source region **203**, the drain region **205** and the first section of the gate region **210** correspond to the source region SP, the drain region DP, and the gate region GP

described with respect to FIG. 1. The active region **202** comprises a source region **204** and a drain region **206** on opposite sides of a second section of the gate region **210** which extends over the active region **202**. The source region **204**, the drain region **206** and the second section of the gate region **210** correspond to the source region SN, the drain region DN, and the gate region GN described with respect to FIG. 1.

[0035] The source regions **203**, **204** are configured to be electrically coupled correspondingly to VDD and VSS power rails on a second side, or a back side, of the substrate. For example, through vias **231**, **232** are arranged to be overlapped correspondingly by the source regions **203**, **204**, and configured to extend through the substrate of an IC device comprising the cell **200**. The through vias **231**, **232** are configured to be in electrical contact with the back sides of the corresponding source regions **203**, **204**, as described herein. A through via is sometimes referred to as a backside via, and is schematically illustrated in the drawings with the label “VB.” The through vias **231**, **232** are configured to electrically couple the corresponding source regions **203**, **204** to corresponding conductive patterns **233**, **234** in a backside-metal (BM) layer which is schematically illustrated in the drawings with the label “BM.” The BM conductive pattern **233** is configured as a VDD power rail on the back side of the substrate, and the BM conductive pattern **234** is configured as a VSS power rail on the back side of the substrate. An example material of the through vias and BM conductive patterns includes metal. Other configurations are within the scopes of various embodiments.

[0036] The first boundary **220** comprises edges **221**, **222**, **223**, **224** connected together to form a closed boundary. In a place-and-route operation (also referred to as “automated placement and routing (APR)”) described herein, cells are placed in an IC layout diagram in abutment with each other at their respective boundaries. For example, the cell **200** is placed in abutment with one or more other cells along the X-X' axis at one or more of the edges **221**, **223**. Additionally or alternatively, the cell **200** is placed in abutment with one or more other cells along the Y-Y' axis at one or more of the edges **222**, **224**. The first boundary **220** is sometimes referred to as “place-and-route boundary” and is schematically illustrated in the drawings with the label “prBoundary.” In the example configuration in FIG. 2A, the first boundary **220** has a rectangular shape, with the edges **221**, **223** parallel to the X-X' axis, and the edges **222**, **224** parallel to the Y-Y' axis. The edge **221** coincides with a top edge of the N well **207** in FIG. 2A. Other configurations are within the scopes of various embodiments. Between the edges **221**, **223** and along the Y axis, the cell **200** contains one PMOS active region, i.e., **201**, and one NMOS active region, i.e., **202**, and is considered to have a height corresponding to one cell height h. In some embodiments, another cell containing along the Y axis two PMOS active regions and two NMOS active regions is considered to have a height corresponding to two cell heights, or double cell height, 2h.

[0037] The cell **200** further comprises dummy gate regions **218**, **219** along the corresponding edges **222**, **224** of the first boundary **220**. In at least one embodiment, centerlines of the dummy gate regions **218**, **219** coincide with the corresponding edges **222**, **224** of the first boundary **220**. Dummy gate regions are schematically illustrated in the drawings with the label “CPODE.” The gate region **210** is an example of “functional gate regions” which, together with the underlying active regions, configure transistors and/or are electrically coupled to one or more other circuit elements. Unlike functional gate regions, dummy gate regions, or non-functional gate regions, are not configured to form transistors together with underlying active regions, and/or one or more transistors formed by dummy gate regions together with the underlying active regions are not electrically coupled to other circuit elements. In at least one embodiment, dummy gate regions include dielectric material in a manufactured IC device. In some embodiments, dummy gate regions and functional gate regions are arranged at the same pitch, i.e., a center-to-center distance, along the X-X' axis. In the example configuration in FIG. 2A, each of the dummy gate regions **218**, **219** has disconnected sections separated from each other along the Y-Y' axis. Other configurations are within the scope of various embodiments. In a place-and-route operation when the cell **200** is placed to abut other cells,

the dummy gate regions **218**, **219** along the edges **222**, **224** of the first boundary **220** are merged with corresponding dummy gate regions of the other cells.

[0038] The described configuration of the cell **200** comprising two active regions **201**, **202** directly adjacent each other along the Y-Y' axis is an example. Other cells in various embodiments include other numbers of active regions arranged along the Y-Y' axis. Two active regions are directly adjacent along the Y-Y' axis when there are no other active regions therebetween. In the example configuration in FIG. 2A, each of the active regions **201**, **202** has, along the X-X' axis, opposite edges (not numbered) coinciding with the edges **222**, **224** of the first boundary **220**. Other configurations are within the scope of various embodiments. The cell **200** comprises a single gate region **210**. This is an example, and other cells in various embodiments include more than one gate regions.

[0039] FIG. 2B is a schematic view showing further layers of the cell **200** above the active regions **201**, **202** and the gate region **210**. Compared to FIG. 2A, the VB through vias **231**, **232** and the BM conductive patterns **233**, **234** are omitted in FIG. 2B, for simplicity.

[0040] In FIG. 2B, the cell **200** further comprises contact structures over and in electrical contact with the corresponding source/drain regions in the active regions **201**, **202**. Contact structures are sometimes referred to as metal-to-device structures, and are schematically illustrated in the drawings with the label "MD." An MD contact structure includes a conductive material formed over a corresponding source/drain region in the corresponding active region to define an electrical connection from one or more devices formed in the active region to other circuitry. In the example configuration in FIG. 2B, MD contact structures **235**, **236** are over and in electrical contact with the corresponding source regions **203**, **204**, and an MD contact structure **237** extends continuously along the Y-Y' axis to be over and in electrical contact with both corresponding drain regions **205**, **206**. The MD contact structure **237** electrically couples the drain regions **205**, **206** together. In some embodiments, MD contact structures and gate regions (including both functional and dummy gate regions) are arranged alternately along the X-X' axis. A pitch, i.e., a center-to-center distance along the X-X' axis, between directly adjacent MD contact structures is the same as the pitch between directly adjacent gate regions. Two gate regions (including functional and/or dummy gate regions) are considered directly adjacent along the X-X' axis where there are no other gate regions (including functional and/or dummy gate regions) therebetween. Two MD conductive patterns are considered directly adjacent along the X-X' axis where there are no other MD conductive patterns therebetween. An example conductive material of MD contact structures includes metal. Other configurations are within the scopes of various embodiments.

[0041] The cell **200** further comprises vias over and in electrical contact with the corresponding gate regions or MD contact structures. A via over and in electrical contact with an MD contact structure is sometimes referred to as via-to-device (VD). A via over and in electrical contact with a gate region is sometimes referred to as via-to-gate (VG). VD and VG vias are schematically illustrated in the drawings with the label "VD/VG." In the example configuration in FIG. 2B, a VG via **238** is over and in electrical contact with the gate region **210**, and a VD via **239** is over and in electrical contact with the MD contact structure **237**. An example material of the VD and VG vias includes metal. Other configurations are within the scopes of various embodiments.

[0042] The cell **200** further comprises one or more metal layers and via layers sequentially and alternately arranged over the VD and VG vias. The lowermost metal layer immediately over and in electrical contact with the VD and VG vias is a metal-zero (M0) layer, a next metal layer immediately over the M0 layer is a metal-one (M1) layer, or the like. A via layer Vn is arranged between and electrically couple the Mn layer and the Mn+1 layer, where n is an integer from zero and up. For example, a via-zero (V0) layer is the lowermost via layer which is arranged between and electrically couple the M0 layer and the M1 layer. Other via layers are V1, V2, or the like.

[0043] The M0 layer is the lowermost metal layer over, or the closest metal layer to, the active regions on the front side of the substrate, as described herein. In some embodiments, all conductive

patterns in the M0 layer belong to the same mask. In at least one embodiment, the conductive patterns in the M0 layer are separated into several masks to meet one or more design and/or manufacturing requirements. At the M0 layer, the cell **200** comprises M0 conductive patterns **241**, **242** correspondingly over the active regions **201**, **202**. In some embodiments, the M0 conductive pattern **241** belongs to one mask, and the M0 conductive pattern **242** belong to another masks. The M0 conductive pattern **241** overlaps and is electrically coupled to the VD via **239**, and the M0 conductive pattern **242** overlaps and is electrically coupled to the VG via **238**. As a result, the M0 conductive pattern **241** is electrically coupled to the drain regions **205**, **206** through the MD contact structure **237** and the VD via **239**, and the M0 conductive pattern **242** is electrically coupled to the gate region **210** through the VG via **238**. The MD conductive patterns **235**, **236** are not electrically coupled to the M0 layer or another metal layer above the M0 layer, and are provided to satisfy one or more design rules, in one or more embodiments. In the example configuration in FIG. 2B, each of the M0 conductive patterns **241**, **242** has opposite edges (not numbered) along the X-X' axis which are spaced from the adjacent dummy gate regions **218**, **219**. In other words, the M0 conductive patterns **241**, **242** are completely arranged inside the first boundary **220**.

[0044] In some embodiments, along the Y-Y' axis and within one cell height h , the cell **200** includes no more than three rows of M0 conductive patterns. In the example configuration in FIG. 2B, the cell **200** includes two rows each extending along the X-X' axis and containing a corresponding one of the M0 conductive patterns **241**, **242**. The two rows are considered directly adjacent along the Y-Y' axis as there are no other rows of M0 conductive patterns therebetween. In at least one embodiment (not shown), an additional row of M0 conductive patterns is arranged along the Y-Y' axis between the two rows containing the M0 conductive patterns **241**, **242**. For example, the additional row of M0 conductive patterns is arranged over a region between the active regions **201**, **202**. In at least one embodiment, by arranging no more than three rows of M0 conductive patterns over one cell height h , it is possible to reduce the area cost of IC devices.

[0045] FIG. 2C is a schematic view showing further layers of the cell **200**. Compared to FIG. 2B, the M0 conductive patterns **241**, **242**, the MD contact structures **235**, **236**, **237**, and the VD/VG vias **238**, **239** are omitted in FIG. 2C, for simplicity.

[0046] The cell **200** further comprises, in the V0 layer over the M0 layer, V0 vias **243**, **244** over and in electrical contact with the corresponding M0 conductive patterns **241**, **242**. The cell **200** further comprises, in the M1 layer over the V0 layer, M1 conductive patterns **245**, **246** over and in electrical contact with the corresponding V0 vias **243**, **244**. As a result, the M1 conductive pattern **245** is electrically coupled to the drain regions **205**, **206** through the MD contact structure **237**, the VD via **239**, the M0 conductive pattern **241** and the V0 vias **243**. The M1 conductive pattern **245** corresponds to the output ZN of the cell **100** described with respect to FIG. 1. The M1 conductive pattern **246** is electrically coupled to the gate region **210** through the VG via **23**, the M0 conductive pattern **242** and the V0 vias **244**. The M1 conductive pattern **246** corresponds to the input I of the cell **100** described with respect to FIG. 1. The M1 conductive patterns **245**, **246** are IO patterns (or IO pins) of the cell **200** that are configured to electrically couple the cell **200** to other circuitry. The M1 layer containing the IO patterns is an example of a pin access layer. Other configurations with other metal layers, such as the M2 layer and/or M3 layer, being the pin access layers are within the scopes of various embodiments. In the example configuration in FIG. 2C, the cell **200** is a simple cell with two IO patterns in a single pin access layer. In some embodiments, a more complex cell has a higher numbers of inputs and/or outputs, resulting in a higher number of IO patterns. It is possible that such complex cells include more than one pin access layers, i.e., have the corresponding IO patterns arranged in more than one metal layers.

[0047] The cell **200** further comprises a second boundary **250** in which IO patterns, such as the M1 conductive patterns **245**, **246**, are arranged. The second boundary **250** comprises edges **251**, **252**, **253**, **254**, **255**, **256** connected together to form a closed boundary. The opposing edges **251**, **254** extend along the X-X' axis and coincide with the corresponding edges **221**, **223** of the first

boundary **220**. The other edges **252**, **253**, **255**, **256** extend obliquely to both the X-X' axis and the Y-Y' axis. For example, the opposing edges **253**, **256** extend along a third axis, e.g., U-U' axis, which is oblique to both the X-X' axis and the Y-Y' axis. The opposing edges **252**, **255** extend along a fourth axis, e.g., V-V' axis, which is oblique to both the X-X' axis and the Y-Y' axis. The U-U' axis is transverse to the V-V' axis. In one or more embodiments, the U-U' axis is perpendicular, to the V-V' axis. In one or more embodiments, the U-U' axis is oblique, i.e., not perpendicular, to the V-V' axis.

[0048] The U-U' axis forms with either of the X-X' axis or the Y-Y' axis an acute angle. For example, as illustrated in FIG. 2C, an angle **257** between the U-U' axis and the Y-Y' axis is an acute angle. The acute angle may be between any one of orientations U, U' and any one of orientations Y, Y'. Similarly, the U-U' axis and the X-X' axis form therebetween an acute angle which may be between any one of orientations U, U' and any one of orientations X, X'. The V-V' axis forms with either of the X-X' axis or the Y-Y' axis an acute angle. For example, the V-V' axis and the Y-Y' axis form therebetween an acute angle **258** which may be between any one of orientations V, V' and any one of orientations Y, Y'. In the example configuration in FIG. 2C, the angles **257**, **258** are the same. The V-V' axis and the X-X' axis form therebetween an acute angle which may be between any one of orientations V, V' and any one of orientations X, X'. Any of the described acute angles, e.g., the angle **257** or **258**, is greater than 0 degrees, and is smaller than 90 degrees. In some embodiments, the acute angle is between 10 degrees and 80 degrees, or between 20 degrees and 70 degrees, or between 30 degrees and 60 degrees, or between 40 degrees and 50 degrees. In at least one embodiment, the acute angle is 45 degrees.

[0049] The IO patterns **245**, **246** are arranged within the second boundary **250**, and also extend obliquely to the X-X' axis and the Y-Y' axis. In the example configuration in FIG. 2C, the IO patterns **245**, **246** are parallel to each other, and also parallel to the edges **253**, **256** of the second boundary **250**. In other words the IO patterns **245**, **246** extend along the U-U' axis. Other configurations where the IO patterns are not necessarily parallel to an edge or edges of the second boundary **250** are within the scopes of various embodiments. In the example configuration in FIG. 2C, the IO patterns **245**, **246** have the same width *w* in a direction perpendicular to a lengthwise direction of the IO patterns **245**, **246**, i.e., in a direction perpendicular to the U-U' axis.

[0050] In some embodiments where IO patterns are arranged in different metal layers or pin access layers, IO patterns in one pin access layer are transverse to IO patterns in another pin access layer. For example, assuming that in addition to the IO patterns **245**, **246** in the M1 layer, the cell **200** has further IO patterns arranged in the M2 layer, the further IO patterns in the M2 layer extend along the V-V' axis to transverse the IO patterns **245**, **246** in the M1 layer, in accordance with some embodiments.

[0051] In some embodiments, conductive patterns in each metal layer below the pin access layer(s), i.e., the metal layer(s) containing IO patterns of a cell, extend along the X-X' axis or the Y-Y' axis. For example, in embodiments where the pin access layer is the M3 layer, the metal layers M0 and M2 have conductive patterns extending along the X-X' axis, whereas the metal layer M1 has conductive patterns extending along the Y-Y' axis. In at least one embodiment, conductive patterns in one or more or all metal layers above the pin access layer(s) extend along the X-X' axis or the Y-Y' axis. For example, when the pin access layer is the M1 layer, for example, as in FIG. 2C, conductive patterns in the M2 layer extend along the X-X' axis in one or more embodiments, or extend along the Y-Y' axis in one or more further embodiments. In some embodiments, each of the IO patterns **245**, **246** is completely arranged within the second boundary **250**. In some embodiments, the pin access layer(s) of a cell is/are the topmost metal layer(s) of the cell. In one or more embodiments, IO patterns of a cell are arranged in non-consecutive metal layers. The described configurations of the IO patterns and the corresponding pin access layer(s) are examples. Other configurations are within the scope of various embodiments.

[0052] As described herein, the cell **200** comprises a first boundary **220** and a second boundary

250. In some embodiments, the first boundary **220** contains therein various layers and features of the cell **200** on the front side of a substrate up to, but not including, a pin access layer. The second boundary **250** contains therein at least IO patterns of the cell **200** in one or more pin access layers. The first boundary **220** and second boundary **250** overlap one another, and have different shapes. A first edge of the first boundary **220** and a second edge of the second boundary **250** overlap each other and form therebetween an acute angle. For example, the edge **222** of the first boundary **220** overlaps the edge **253** of the second boundary **250**, and an acute angle corresponding to the angle **257** is formed between the edge **222** of the first boundary **220** and the edge **253** of the second boundary **250**. For another example, the edge **222** of the first boundary **220** overlaps the edge **252** of the second boundary **250**, and an acute angle corresponding to the angle **258** is formed between the edge **222** of the first boundary **220** and the edge **252** of the second boundary **250**. As shown in FIG. 2C, the dummy gate region **219** arranged along the edge **222** of the first boundary **220** extends from inside the second boundary **250**, across the edge **252** or **253** of the second boundary **250**, to outside the second boundary **250**. The dummy gate region **218** arranged along the edge **224** of the first boundary **220** extends from inside the second boundary **250**, across the edge **255** or **256** of the second boundary **250**, to outside the second boundary **250**.

[0053] The described and illustrated shape(s) of the first boundary **220** and/or the second boundary **250** is/are examples. Other configurations are within the scopes of various embodiments. For example, in at least one embodiment, the first boundary **220** has a shape other than a rectangular shape. For further examples, although the second boundary **250** is described to have a shape of a convex hexagon, in one or more embodiments, the second boundary **250** has a shape of a concave hexagon, as described herein.

[0054] In some embodiments, in a place-and-route operation as described herein, when the cell **200** is placed in abutment with a further cell so that the first boundary **220** of the cell **200** abuts a corresponding first boundary of the further cell, the second boundary **250** of the cell **200** also abuts, at the same time, a corresponding second boundary of the further cell.

[0055] In some embodiments, various layers and features on the front side of a substrate are manufactured in a front-end-of-line (FEOL) stage, and are sometimes referred to as FEOL layers and features. Layers and features other than the FEOL layers and features, are manufactured in a back-end-of-line (BEOL) stage, and are sometimes referred to as BEOL layers and features. Example FEOL layers and features include, but are not limited to, N- and P-wells, active regions, gate regions including both functional and dummy gate regions, MD contact structures, VD and VG vias, and M0 conductive patterns. FEOL layers and features are arranged inside a first boundary, such as the first boundary **220**. Example BEOL layers and features include, but are not limited to, metal layers and via layers above the FEOL layers and features, as well as metal layers and via layers on the back side of the substrate. Pin access layers of cells are examples of BEOL layers and are arranged inside a second boundary, such as the second boundary **250**. In the example configuration in FIGS. 2A-2C, the schematic views in FIGS. 2A-2B show FEOL layers and features of the cell **200**, and the schematic view in FIG. 2C show BEOL layers and features of the cell **200**, in accordance with some embodiments.

[0056] FIG. 2D is a schematic view of the layout diagram of the cell **200** including the layers in FIGS. 2A-2C, in accordance with some embodiments. The schematic view in FIG. 2D is a combination of the schematic views in FIGS. 2A-2C. For simplicity, the first boundary **220** and the VB via **231** are omitted in FIG. 2D.

[0057] In other approaches with IO patterns parallel to active regions or gate regions, pin access during APR is a design consideration, especially as ICs are increasingly miniaturized at greater scales. For example, one or more access points for accessing a pin or IO pattern of a cell are not usable due to existing metal tracks and/or due to one or more other neighboring cells blocking the access points. An attempt to solve this issue includes increasing the number of M0 tracks, or rows of M0 conductive patterns, in the cell; however, such attempt increases the cell height which, in

turn, potentially leads to degradation in one or more of performance, power and area (PPA).

[0058] In some embodiments, by arranging the IO patterns at an acute angle to the gate regions, it is possible to provide more access points to the IO patterns without having to increase the number of M0 tracks in a cell. As a result, pin access challenges are resolvable, while maintaining or even achieving PPA improvements, in at least one embodiment. As described herein, in some embodiments, there are no more than three M0 tracks over a cell height which contributes to reduction of the area cost, with further associated advantages. In at least one embodiment, by providing a different boundary for at least oblique IO patterns, a place-and-route operation for cells having oblique IO patterns is facilitated.

[0059] FIG. 3 is a schematic cross-sectional view, taken along lines III-III in FIGS. 2A-2D, of an IC device 300, in accordance with some embodiments. The IC device 300 comprises, in FIG. 3, an inverter corresponding to the cell 200 described with respect to FIGS. 2A-2D. Components in FIG. 3 having corresponding components in FIGS. 2A-2D are designated by the reference numerals of FIGS. 2A-2D increased by 100. For example, the MD contact structure 236 in FIG. 2B corresponding to an MD contact structure 336 in FIG. 3.

[0060] As shown in FIG. 3, the IC device 300 comprises a substrate 360 over which circuit elements and structures corresponding to the cell 200 are formed. The substrate 360 has a first side 361 and a second side 362 opposite one another along a thickness direction of the substrate 360, i.e., along a Z-Z' axis. In at least one embodiment, the first side 361 is referred to as “upper side” or “front side” or “device side,” whereas the second side 362 is referred to as “lower side” or “back side.” The substrate comprises, in at least one embodiment, silicon, silicon germanium (SiGe), gallium arsenic, or other suitable semiconductor or dielectric materials.

[0061] N-type and P-type dopants are added to the front side 361 of the substrate 360 to correspondingly form N wells in an NMOS active region, and P wells in a PMOS active region. For example, N wells 304, 306 are formed over the front side 361 of the substrate 360, and correspond to the source region 204 and drain region 206 in FIGS. 2A-2B. The N wells 304, 306 configure source/drain regions of a transistor 3NM corresponding to the transistor NM in FIG. 2A. In some embodiments, isolation structures are formed between adjacent P wells and N wells. For simplicity, isolation structures are omitted from FIG. 3. A gate of the transistor 3NM comprises a stack of gate dielectric layers 363, 364, and a gate electrode 310. Similar stacks of gate dielectric layers 363, 364 are formed under dummy gate electrodes 318, 319 on opposite sides of the gate electrode 310 along the X-X' axis. In at least one embodiment, each of the described stack of gate dielectric layers is replaced with a gate dielectric layer instead of multiple gate dielectrics. Example materials of the gate dielectric layer or layers include HfO₂, ZrO₂, or the like. Example materials of the gate electrodes include polysilicon, metal, or the like. In some embodiments, the dummy gate electrodes 318, 319 include dielectric materials.

[0062] The IC device 300 further comprises MD contact structures for electrically coupling source/drains of transistors to other circuit elements. In FIG. 3, MD contact structures comprise MD contact structures 336, 337 correspondingly over and in electrical contact with the N wells or source/drain regions 304, 306. Further, VD vias (not shown in FIG. 3) and VG vias are correspondingly over and in electrical contact with the MD contact structures and gate electrodes. For example, as shown in FIG. 3, a VG via 338 corresponding to the VG via 238 is over and in electrical contact with the gate electrode 310.

[0063] The IC device 300 further comprises an interconnect structure 368 which is over the VD and VG vias, and comprises a plurality of metal layers M0, M1, . . . and a plurality of via layers V0, V1, . . . arranged alternately in the thickness direction of the substrate 360, i.e., along the Z-Z' axis. The interconnect structure 368 further comprises various interlayer dielectric (ILD) layers (not shown) in which the metal layers and via layers are embedded. The metal layers and via layers of the interconnect structure 368 are configured to electrically couple various elements or circuits of the IC device 300 with each other, and with external circuitry. For simplicity, metal layers and

via layers above the M1 layer are omitted in FIG. 3. The M0 layer comprises an M0 conductive pattern **342** corresponding to the M0 conductive pattern **242** in FIGS. 2B-2C, over and in electrical contact with the VG via **338**. The V0 layer comprises a V0 via **344** corresponding to the V0 via **244** in FIG. 2C, over and in electrical contact with the M0 conductive pattern **342**. The M1 layer comprises an M1 conductive pattern **346** corresponding to the M1 conductive pattern **246** in FIG. 2C, over and in electrical contact with the V0 via **344**. The M1 conductive pattern **346** is an IO pattern or IO pin corresponding to the input of the inverter in the cell **200**.

[0064] The IC device **300** further comprises a VB through via **332** corresponding to the VB through via **232** in FIG. 2A. The VB through via **332** extends from the back side **362** of the substrate **360** towards the front side **361**, to be in electrical contact with a back side **365** of the N well **304**.

[0065] The IC device **300** further comprises a backside-metal layer, such as a backside-metal-zero (BM0) layer under the back side **362** of the substrate **360**. On the back side **362** of the substrate **360**, the BM0 layer is the uppermost metal layer under, or the closest metal layer to, the active regions or source/drains of the transistors of the IC device **300**. In the example configuration in FIG. 3, the BM0 layer comprises a conductive pattern **334** under and in electrical contact with the VB through via **332**. The conductive pattern **334** is configured as a VSS power rail and corresponds to the BM conductive pattern **234** in FIG. 2A. Other conductive patterns (not shown) of the BM0 layer are configured to as VDD or VSS power rails for delivery of power supply or ground voltages to the circuitry of the IC device **300**. In at least one embodiment, the IC device **300** comprises one or more further via layers, dielectric layers and metal layers (not shown) under the BM0 layer to form interconnections among circuit elements of the IC device **300** and/or to form electrical connections to external circuitry. Via layers and metal layers from the BM0 layer and below are sometimes referred to as backside via layers and metal layers. For simplicity, dielectric layers, and backside via and metal layers lower than the BM0 layer are omitted from FIG. 3. In at least one embodiment, one or more advantages described herein are achievable in the IC device **300**.

[0066] FIG. 4 is a circuit diagram of a cell **400**, in accordance with some embodiments. In FIG. 4, the cell **400** is a two-input NAND gate. This is an example, and other cells are within the scope of various embodiments.

[0067] In the example circuit diagram in FIG. 4, the cell **400** comprises PMOS transistors TPA1, TPA2 and NMOS transistors TNA1, TNA2. The transistors TPA1, TPA2 are coupled in parallel between VDD and an output ZN. The transistors TNA1, TNA2 are coupled in series between the output ZN and VSS. Gates of the transistors TPA1, TNA1 are commonly coupled to a first input A1. Gates of the transistors TPA2, TNA2 are commonly coupled to a second input A2. The cell **400** is configured to perform a NAND logic operation on input signals at the inputs A1, A2, and generate an output signal at the output ZN. In at least one embodiment, the described NMOS transistors are replaced with NMOS transistors, and vice versa.

[0068] FIGS. 5A-5C are schematic views of layout diagrams of corresponding cells **500A**, **500B**, **500C**, in accordance with some embodiments. In at least one embodiment, the layout diagrams of one or more of the cells **500A-500C** are stored as standard cells in a standard cell library on a non-transitory computer-readable recording medium. The cells **500A-500C** are two-input NAND gates corresponding to the cell **400**. In the example configuration in FIGS. 5A-5C, the cells **500A-500C** are two-input NAND gates with a driving strength of 1, also referred to as “ND2D1.” Other example two-input NAND gates with different driving strengths are within the scopes of various embodiments.

[0069] In FIG. 5A, the cell **500A** comprises active regions **501**, **502**, functional gate regions **510**, **511**, dummy gate regions **518**, **519**. The active regions **501**, **502** extend along the X-X' axis. The active region **501** is over an N well **507**. The gate region **510** extends across the active regions **501**, **502** along the Y-Y' axis and configure, together with the active regions **501**, **502**, transistors TPA1, TNA1 as described with respect to FIG. 4. The gate region **511** extends across the active regions

501, **502** along the Y-Y' axis and configure, together with the active regions **501**, **502**, transistors TPA2, TNA2 as described with respect to FIG. 4. A source/drain region of the transistor TPA1 and a source/drain region of the transistor TPA2 are formed in the active region **501** are commonly electrically coupled to a back side VDD power rail **533**, by a VB through via VB51. A source/drain region of the transistor TNA2 is electrically coupled to a back side VSS power rail **534**, by another VB through via (not shown).

[0070] The cell **500A** further comprises MD contact structures MD51-MD55, VD vias VD51, VD52, VG vias VG51, VG52, M0 conductive patterns M051-M053, V0 vias V051-V053, and M1 conductive patterns M151-M153. The MD contact structure MD51 is over and in electrical contact with the corresponding source/drain regions of the transistors TPA1, TNA1 in the corresponding active regions **501**, **502**, and is further electrically coupled to the VD via VD51, then to the M0 conductive pattern M051, then to the V0 via V053, then to the M1 conductive pattern M153. The MD contact structure MD52 is over and in electrical contact with corresponding source/drain regions of the transistors TPA1, TPA2, and is not further electrically coupled to other circuit elements. The MD contact structure MD53 is over and in electrical contact with corresponding source/drain regions of the transistors TNA1, TNA2, and is not further electrically coupled to other circuit elements. The MD contact structure MD54 is over and in electrical contact with a corresponding source/drain region of the transistor TPA2, and is further electrically coupled to the VD via VD52, then to the M0 conductive pattern M051, then to the V0 via V053, then to the M1 conductive pattern M153. The MD contact structure MD55 is over and in electrical contact with a corresponding source/drain region of the transistor TNA2, and is not further electrically coupled to other circuit elements. The gate region **510** is the common gate region to the transistors TPA1, TNA1, and is electrically coupled to the VG via VG51, then to the M0 conductive pattern M052, then to the V0 via V051, then to the M1 conductive pattern M151. The gate region **511** is the common gate region to the transistors TPA2, TNA2, and is electrically coupled to the VG via VG52, then to the M0 conductive pattern M053, then to the V0 via V052, then to the M1 conductive pattern M152.

[0071] The M1 conductive patterns M151, M152, M153 are IO patterns of the cell **500A** and correspond to the inputs A1, A2 and the output ZN described with respect to FIG. 4. The M1 layer is, therefore, the pin access layer of the cell **500A**. The IO patterns M151, M152, M153 extend along the U-U' axis and form an acute angle with the gate regions **510**, **511**, as well as the dummy gate regions **518**, **519**. The IO patterns M151, M152, M153 are arranged in a boundary **550** having edges **551**, **552**, **553**, **554**, **555**, **556** connected together to form a closed boundary. The opposing edges **551**, **554** extend along the X-X' axis. The other edges **552**, **553**, **555**, **556** extend obliquely to both the X-X' axis and the Y-Y' axis. For example, the opposing edges **553**, **556** extend along the U-U' axis, and the opposing edges **552**, **555** extend along the V-V' axis. In the example configuration in FIG. 5A, the IO patterns M151, M152, M153 are parallel to each other, and also parallel to the edges **553**, **556** of the boundary **550**. The IO patterns M151, M152, M153 are arranged at a pitch p in a direction perpendicular to a lengthwise direction of the IO patterns M151, M152, M153, i.e., in a direction perpendicular to the U-U' axis. In the example configuration in FIG. 5A, the IO patterns M151, M153 are spaced from each other by one pitch p. The IO patterns M151, M152 are spaced from each other by two pitches, i.e., 2p, meaning that an additional M1 conductive pattern is placeable between the IO patterns M151, M152. The boundary **550** has a shape of a convex hexagon.

[0072] Besides the boundary **550**, which is hexagonal, contains the IO patterns and corresponds to the second boundary **250**, the cell **500A** further comprises a further boundary corresponding to the first boundary **220**. The further boundary, also referred to as the first boundary of the cell **500A**, is rectangular, and has two opposing edges defined by the dummy gate regions **518**, **519**, and two further opposing edges coinciding with the edges **551**, **554** of the boundary **550**. For simplicity, the first, rectangular boundary of the cell **500A** is omitted in FIG. 5A. The first boundary contains

therein various layers and features of the cell **500A** on the front side of a substrate up to, but not including, the pin access layer (i.e., the M1 layer).

[0073] The cell **500A** has a height along the Y-Y' axis corresponding to one cell height *h*. Similar to the cell **200**, the cell **500A** includes, in one or more embodiments, no more than three rows of M0 conductive patterns over the one cell height *h*. In the example configuration in FIG. 5A, the cell **500A** includes two rows of M0 conductive patterns, the first row containing the M0 conductive pattern **M051**, and the second row containing the M0 conductive patterns **M052**, **M053**. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell **500A**.

[0074] In FIG. 5B, the cell **500B** is similar to the cell **500A** with respect to the first boundary and the layers and features in the first boundary, at and below the M0 layer. The cell **500B** is different from the cell **500A** in the configuration of a second boundary and at least one of the IO patterns. The cell **500B** has a second boundary **560** with edges **561-566**. The second boundary **560** contains therein the IO patterns **M151**, **M152**, **M154**. Unlike the boundary **550** which has a shape of a convex hexagon, the second boundary **560** has a shape of a concave hexagon. Specifically, an internal angle at a corner **567** between the edges **565**, **566** of the boundary **560** is greater than 180 degrees, whereas internal angles at the other five corners of the boundary **560** are smaller than 180 degrees. Due to the different shape of the boundary **560**, the IO pattern corresponding to the output ZN, i.e., an M1 conductive pattern **M154** is shorter than the corresponding M1 conductive pattern **M153** in the cell **500A**. Further, a location of a V0 via **V054** coupled to the M1 conductive pattern **M154** is different from that of the corresponding V0 via **V053** in the cell **500A**. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell **500B**.

[0075] In FIG. 5C, the cell **500C** is similar to the cells **500A**, **500B** with respect to the first boundary and the layers and features in the first boundary, at and below the M0 layer. The cell **500C** is different from the cells **500A**, **500B** in the configuration of a second boundary and at least one of the IO patterns. The cell **500C** has a second boundary **570** with edges **571-576**. The second boundary **560** contains therein the IO patterns **M151**, **M154**, **M155**. Unlike the boundary **550** which has a shape of a convex hexagon, the second boundary **570** has a shape of a concave hexagon. Unlike the boundary **560** which has a shape of a concave hexagon with one internal corner greater than 180 degrees, the second boundary **570** has a shape of a concave hexagon with two internal corners greater than 180 degrees. Specifically, an internal angle at a corner **577** between the edges **575**, **576** of the boundary **560** is greater than 180 degrees, and an internal angle at a corner **578** between the edges **572**, **573** of the boundary **560** is also greater than 180 degrees. Internal angles at the other four corners of the boundary **570** are smaller than 180 degrees. Due to the different shape of the boundary **570**, the IO pattern corresponding to the input A2, i.e., an M1 conductive pattern **M155** is shorter than the corresponding M1 conductive pattern **M152** in the cells **500A**, **500B**. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell **500C**.

[0076] FIGS. 6A-6C are schematic views of layout diagrams of corresponding cells **600A**, **600B**, **600C**, in accordance with some embodiments. In at least one embodiment, the layout diagrams of one or more of the cells **600A-600C** are stored as standard cells in a standard cell library on a non-transitory computer-readable recording medium. The cells **600A-600C** are inverters corresponding to the cell **100**.

[0077] In FIG. 6A, the cell **600A** is an inverter with a driving strength of 4, also referred to as "INVD4."

[0078] The cell **600A** comprises active regions **601**, **602** extending along the X-X' axis. The active region **601** is over an N well **607**. Unlike the cell **200** with one functional gate region **210** over the active regions **201**, **202**, the cell **600A** comprises four functional gate regions **610-613** over the active regions **601**, **602**. All four PMOS transistors configured by the four functional gate regions

610-613 and the active region **601** have corresponding gate regions electrically coupled together, corresponding source regions coupled together, and corresponding drain regions coupled together. As a result, all four PMOS transistors are coupled to function as one PMOS transistor corresponding to the transistor PM in FIG. 1, but with a drive strength of 4. Similarly, all four NMOS transistors configured by the four functional gate regions **610-613** and the active region **602** have corresponding gate regions electrically coupled together, corresponding source regions coupled together, and corresponding drain regions coupled together. As a result, all four NMOS transistors are coupled to function as one NMOS transistor corresponding to the transistor NM in FIG. 1, but with a drive strength of 4. The PMOS and NMOS transistors of the cell **600A** are electrically coupled together by various MD contact structures, VG vias, VD vias, M0 conductive patterns, V0 vias and M1 conductive patterns in a manner similar to the cell **200** described with respect to FIGS. 2A-2D. The PMOS and NMOS transistors of the cell **600A** are further electrically coupled to back side VDD and VSS power rails by various VB through vias in a manner similar to the cell **200** described with respect to FIGS. 2A-2D.

[0079] The cell **600A** comprises an M1 conductive pattern **645** corresponding to the M1 conductive pattern **245** of the cell **200**, and corresponding to the output ZN of the inverter. The cell **600A** further comprises an M1 conductive pattern **646** corresponding to the M1 conductive pattern **246** of the cell **200**, and corresponding to the input I of the inverter. The M1 conductive patterns **645**, **646** are IO patterns of the cell **600A**. The M1 layer is, therefore, the pin access layer of the cell **600A**. The IO patterns **645**, **646** extend along the U-U' axis and form an acute angle with the gate regions **610-613**, as well as the dummy gate regions **618**, **619**. The IO patterns **645**, **646** are arranged in a boundary **650** having edges **651**, **652**, **653**, **654**, **655**, **656** connected together to form a closed boundary. The opposing edges **651**, **654** extend along the X-X' axis. The other edges **652**, **653**, **655**, **656** extend obliquely to both the X-X' axis and the Y-Y' axis. For example, the opposing edges **653**, **656** extend along the U-U' axis, and the opposing edges **652**, **655** extend along the V-V' axis. In the example configuration in FIG. 6A, the IO patterns **645**, **646** are parallel to each other, and also parallel to the edges **653**, **656** of the boundary **650**. The boundary **650** has a shape of a convex hexagon. In some embodiments, the boundary **650** has a shape of a concave hexagon as described with respect to FIG. 5B or FIG. 5C.

[0080] Besides the boundary **650**, which is hexagonal, contains the IO patterns and corresponds to the second boundary **250**, the cell **600A** further comprises a further boundary corresponding to the first boundary **220**. The further boundary, also referred to as the first boundary of the cell **600A**, is rectangular, and has two opposing edges defined by the dummy gate regions **618**, **619**, and two further opposing edges coinciding with the edges **651**, **654** of the boundary **650**. For simplicity, the first, rectangular boundary of the cell **600A** is omitted in FIG. 6A. The first boundary contains therein various layers and features of the cell **600A** on the front side of a substrate up to, but not including, the pin access layer (i.e., the M1 layer).

[0081] The cell **600A** has a height along the Y-Y' axis corresponding to one cell height h. Similar to the cell **200**, the cell **600A** includes, in one or more embodiments, no more than three rows of M0 conductive patterns over the one cell height h. In the example configuration in FIG. 6A, the cell **600A** includes two rows of M0 conductive patterns. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell **600A**.

[0082] In FIG. 6B, the cell **600B** is an inverter with a driving strength of 4, i.e., the cell **600B** is an INVD4 cell. A difference between the INVD4 cells **600A** and **600B** is that the cell **600B** has a double cell height, i.e., 2h, whereas the cell **600A** has a single cell height h.

[0083] The cell **600B** comprises active regions **621-624** extending along the X-X' axis. The active region **621** is over an N well **627**, and the active region **623** is over an N well **628**. The cell **600B** comprises two functional gate regions **630**, **631** over the active regions **621-624**. All four PMOS transistors configured by the two functional gate regions **630**, **631** and the two active regions **621**, **623** have corresponding gate regions electrically coupled together, corresponding source regions

coupled together, and corresponding drain regions coupled together. As a result, all four PMOS transistors are coupled to function as one PMOS transistor corresponding to the transistor PM in FIG. 1, but with a drive strength of 4. Similarly, all four NMOS transistors configured by the two functional gate regions **630**, **631** and the two active region **622**, **624** have corresponding gate regions electrically coupled together, corresponding source regions coupled together, and corresponding drain regions coupled together. As a result, all four NMOS transistors are coupled to function as one NMOS transistor corresponding to the transistor NM in FIG. 1, but with a drive strength of 4. The PMOS and NMOS transistors of the cell **600B** are electrically coupled together by various MD contact structures, VG vias, VD vias, M0 conductive patterns, V0 vias and M1 conductive patterns in a manner similar to the cell **200** described with respect to FIGS. 2A-2D. The PMOS and NMOS transistors of the cell **600B** are further electrically coupled to back side VDD and VSS power rails by various VB through vias in a manner similar to the cell **200** described with respect to FIGS. 2A-2D.

[0084] The cell **600B** comprises an M1 conductive pattern **635** corresponding to the M1 conductive pattern **245** of the cell **200**, and corresponding to the output ZN of the inverter. The cell **600B** further comprises an M1 conductive pattern **636** corresponding to the M1 conductive pattern **246** of the cell **200**, and corresponding to the input I of the inverter. The M1 conductive patterns **635**, **636** are IO patterns of the cell **600B**. The M1 layer is, therefore, the pin access layer of the cell **600B**. The IO patterns **635**, **636** extend along the U-U' axis and form an acute angle with the gate regions **630**, **631**, as well as the dummy gate regions **638**, **639**. The IO patterns **635**, **636** are arranged in a boundary **660** having edges **661-670** connected together to form a closed boundary. The edges **661**, **666** extend along the X-X' axis. The other edges **662-665** and **667-670** extend obliquely to both the X-X' axis and the Y-Y' axis. For example, the edges **663**, **665**, **668**, **670** extend along the U-U' axis, and the edges **662**, **664**, **667**, **669** extend along the V-V' axis. In the example configuration in FIG. 6B, the IO patterns **635**, **636** are parallel to each other, and also parallel to the edges **663**, **665**, **668**, **670** of the boundary **660**.

[0085] The boundary **660** has a shape being a combination of two hexagons. For example, the first hexagon is defined by the edges **661**, **662**, **663**, a line **659**, and the edges **669**, **670**. The second hexagon is defined by the line **659**, and the edges **664**, **665**, **666**, **667**, **668**. In the example configuration in FIG. 6B, both the first and second hexagons are convex hexagons. In some embodiments, at least one of the first and second hexagons is a concave hexagon as described with respect to FIG. 5B or FIG. 5C. Further, the arrangement of the first and second hexagons in FIG. 6B as being stacked or combined along the Y-Y' axis is an example. In at least one embodiment, the first and second hexagons in a cell are stacked or combined along the X-X' axis. In at least one embodiment, more than two hexagons are combined in a cell.

[0086] Besides the boundary **660**, which contains the IO patterns and corresponds to the second boundary **250**, the cell **600B** further comprises a further boundary corresponding to the first boundary **220**. The further boundary, also referred to as the first boundary of the cell **600B**, is rectangular, and has two opposing edges defined by the dummy gate regions **638**, **639**, and two further opposing edges coinciding with the edges **661**, **666** of the boundary **660**. For simplicity, the first, rectangular boundary of the cell **600B** is omitted in FIG. 6B. The first boundary contains therein various layers and features of the cell **600B** on the front side of a substrate up to, but not including, the pin access layer (i.e., the M1 layer).

[0087] The cell **600B** includes four active regions **621-624** arranged consecutively along the Y-Y' axis, and has a height along the Y-Y' axis corresponding to 2h. Similar to the cell **200**, the cell **600B** includes, in one or more embodiments, no more than three rows of M0 conductive patterns over each cell height h. In the example configuration in FIG. 6B, the cell **600B** includes four rows of M0 conductive patterns. In some embodiments, a maximum number of rows of M0 conductive patterns in the cell **600B** is six. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell **600B**.

[0088] In FIG. 6C, the cell **600C** is an inverter with a driving strength of 6, also referred to as “INVD6.” The cell **600C** also has a double cell height, i.e., $2h$.

[0089] The cell **600C** comprises active regions **671-674** extending along the X-X' axis. The active region **671** is over an N well **616**, and the active region **673** is over an N well **617**. The cell **600C** comprises four functional gate regions **675-678** over the active regions **671-674**. Two functional gate regions **676, 677** and the active region **671** configure two PMOS transistors, and the four functional gate regions **675-678** and the active region **673** configure four further PMOS transistors. All six PMOS transistors have corresponding gate regions electrically coupled together, corresponding source regions coupled together, and corresponding drain regions coupled together. As a result, all six PMOS transistors are coupled to function as one PMOS transistor corresponding to the transistor PM in FIG. 1, but with a drive strength of 6. Similarly, all six NMOS transistors configured by the functional gate regions **675-678** and the two active region **672, 674** have corresponding gate regions electrically coupled together, corresponding source regions coupled together, and corresponding drain regions coupled together. As a result, all six NMOS transistors are coupled to function as one NMOS transistor corresponding to the transistor NM in FIG. 1, but with a drive strength of 6. The PMOS and NMOS transistors of the cell **600C** are electrically coupled together by various MD contact structures, VG vias, VD vias, M0 conductive patterns, V0 vias and M1 conductive patterns in a manner similar to the cell **200** described with respect to FIGS. 2A-2D. The PMOS and NMOS transistors of the cell **600C** are further electrically coupled to back side VDD and VSS power rails by various VB through vias in a manner similar to the cell **200** described with respect to FIGS. 2A-2D.

[0090] The cell **600C** comprises an M1 conductive pattern **657** corresponding to the M1 conductive pattern **245** of the cell **200**, and corresponding to the output ZN of the inverter. The cell **600C** further comprises an M1 conductive pattern **658** corresponding to the M1 conductive pattern **246** of the cell **200**, and corresponding to the input I of the inverter. The M1 conductive patterns **657, 658** are IO patterns of the cell **600C**. The M1 layer is, therefore, the pin access layer of the cell **600C**. The IO patterns **657, 658** extend along the U-U' axis and form an acute angle with the gate regions **675-678**, as well as the dummy gate regions **603-606** and **608-609**. The IO patterns **657, 658** are arranged in a boundary **680** having edges **681-686** connected together to form a closed, hexagonal boundary. The edges **681, 684** extend along the X-X' axis. The other edges extend obliquely to both the X-X' axis and the Y-Y' axis. For example, the edges **683, 686** extend along the U-U' axis, and the edges **682, 685** extend along the V-V' axis. In the example configuration in FIG. 6C, the IO patterns **657, 658** are parallel to each other, and also parallel to the edges **683, 686** of the boundary **680**. In some embodiments, the boundary **680** has a shape of a concave hexagon as described with respect to FIG. 5B or FIG. 5C.

[0091] Besides the boundary **680**, which contains the IO patterns and corresponds to the second boundary **250**, the cell **600C** further comprises a further boundary corresponding to the first boundary **220**. The further boundary, also referred to as the first boundary **690** of the cell **600C**, has a shape being a combination of three rectangles.

[0092] FIG. 6D which is a schematic view showing the boundaries **680, 690** of the cell **600C**. The first boundary **690** of the cell **600C** includes edges **687-689** and **691-699** connected together to form a closed boundary. The edges **687, 689, 692, 694, 696, 698** in FIG. 6D correspond to dummy gate regions **603-606** and **608-609** in FIG. 6C. The edges **688, 693** in FIG. 6D correspond to a lower edge of the N well **616** in FIG. 6C. The edge **691** in FIG. 6D coincides with the edge **681** in FIG. 6C. The edges **695, 699** in FIG. 6D correspond to a lower edge of the N well **617** in FIG. 6C. The edge **697** in FIG. 6D coincides with the edge **684** in FIG. 6C. The first boundary **690** contains therein various layers and features of the cell **600C** on the front side of a substrate up to, but not including, the pin access layer (i.e., the M1 layer).

[0093] The cell **600C** includes four active regions **671-674** arranged consecutively along the Y-Y' axis, and has a height along the Y-Y' axis corresponding to $2h$. Similar to the cell **200**, the cell

600C includes, in one or more embodiments, no more than three rows of M0 conductive patterns over each cell height *h*. In the example configuration in FIG. 6C, the cell **600C** includes four rows of M0 conductive patterns. In some embodiments, a maximum number of rows of M0 conductive patterns in the cell **600C** is six. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell **600C**.

[0094] FIG. 7 is a schematic circuit diagram of a cell **700** of an IC device, in accordance with some embodiments. In the example configuration in FIG. 7, the cell **700** comprises an AND-OR-Invert (AOI) logic with two 2-input AND gates corresponding to a standard cell sometimes referred to as an AOI22D1 cell.

[0095] The cell **700** comprises inputs A1, A2, B1, B2, an output ZN, and a plurality of transistors PA1, PA2, PB1, PB2, NA1, NA2, NB1, NB2 electrically coupled together to perform, in operation, a predetermined function of the cell **700**. In the example configuration in FIG. 7, the cell **700** comprises PMOS transistors PA1, PA2, PB1, PB2 and NMOS transistors NA1, NA2, NB1, NB2. Gates of the transistors PA1, NA1 are electrically coupled to the input A1. Gates of the transistors PA2, NA2 are electrically coupled to the input A2. Gates of the transistors PB1, NB1 are electrically coupled to the input B1. Gates of the transistors PB2, NB2 are electrically coupled to the input B2. Sources of the transistors PB1, PB2 are electrically coupled to VDD. Drains of the transistors PB1, PB2 are electrically coupled to a node C. As a result, the transistors PB1, PB2 are electrically coupled in parallel between VDD and the node C. Sources of the transistors PA1, PA2 are electrically coupled to the node C. Drains of the transistors PA1, PA2 are electrically coupled to the output ZN. As a result, the transistors PA1, PA2 are electrically coupled in parallel between the node C and the output ZN. The parallel coupled transistors PB1, PB2 and the parallel coupled transistors PA1, PA2 are electrically coupled in series at the node C. Sources of the transistors NA2, NB2 are electrically coupled to VSS. A drain of the transistor NA2 is electrically coupled to a source of the transistor NA1 at a node D. As a result, the transistors NA1, NA2 are electrically coupled in series. A drain of the transistor NB2 is electrically coupled to a source of the transistor NB1 at a node E. As a result, the transistors NB1, NB2 are electrically coupled in series. Drains of the transistors NA1, NB1 are electrically coupled to the output ZN. As a result, the serially coupled transistors NA1, NA2 and the serially coupled transistors NB1, NB2 are coupled in parallel between the output ZN and VSS.

[0096] FIGS. 8A-8B are schematic views of layout diagrams of corresponding cells **800A**, **800B** in accordance with some embodiments. In at least one embodiment, the layout diagrams of one or more of the cells **800A**, **800B** are stored as standard cells in a standard cell library on a non-transitory computer-readable recording medium. The cells **800A**, **800B** are AOI22D1 cells corresponding to the cell **700**. In the example configuration in FIGS. 8A-8B, the cells **800A**, **800B** have a driving strength of 1. Other cells with different driving strengths are within the scopes of various embodiments.

[0097] In FIG. 8A, the cell **800A** comprises active regions **821-824** extending along the X-X' axis. The active regions **822**, **823** are directly adjacent to each other and are over an N well **827**. The cell **800A** comprises functional gate regions **830**, **831**, **832**, **833** over the active regions **821-824**. The gate regions **830**, **832** are aligned along the Y-Y' axis and are electrically isolated from each other. The gate regions **831**, **833** are aligned along the Y-Y' axis and are electrically isolated from each other. The gate regions **830**, **831** configure, together with the active region **821**, transistors NB2, NB1 as described with respect to FIG. 7. The functional gate regions **830**, **831** configure, together with the active region **822**, transistors PB2, PB1 as described with respect to FIG. 7. The gate regions **832**, **833** configure, together with the active region **823**, transistors PA2, PA1 as described with respect to FIG. 7. The functional gate regions **832**, **833** configure, together with the active region **824**, transistors NA2, NA1 as described with respect to FIG. 7. The PMOS and NMOS transistors of the cell **800A** are electrically coupled together by various MD contact structures, VG vias, VD vias, M0 conductive patterns, V0 vias and M1 conductive patterns to realize the circuitry

described with respect to FIG. 7. For example, an elongated MD contact structure MD81 extends across two active regions 822, 823, electrically couples the corresponding source/drain regions of the transistors PB2, PB1, PA2, PA1, and corresponds to node C in FIG. 7. The PMOS and NMOS transistors of the cell 800A are further electrically coupled to back side VDD and VSS power rails by various VB through vias to realize the circuitry described with respect to FIG. 7.

[0098] The cell 800A comprises M1 conductive patterns M181-M185 corresponding to the inputs A1, A2, B1, B2 and the output ZN described with respect to FIG. 7. The M1 conductive patterns M181-M185 are IO patterns of the cell 800A. The M1 layer is, therefore, the pin access layer of the cell 800A. The IO patterns M181-M185 extend along the U-U' axis and form an acute angle with the gate regions 830-833, as well as the dummy gate regions 838, 839. The IO patterns M181-M185 are arranged in a boundary 860 which has a shape being a combination of two hexagons similar to the boundary 660 described with respect to FIG. 6B.

[0099] Besides the boundary 860, which contains the IO patterns and corresponds to the second boundary 250, the cell 800A further comprises a further boundary corresponding to the first boundary 220. The further boundary, also referred to as the first boundary of the cell 800A, is rectangular, and has two opposing edges defined by the dummy gate regions 838, 839, and two further opposing edges coinciding with the uppermost and lowermost edges 861, 866 of the boundary 860. For simplicity, the first, rectangular boundary of the cell 800A is omitted in FIG. 8A. The first boundary contains therein various layers and features of the cell 800A on the front side of a substrate up to, but not including, the pin access layer (i.e., the M1 layer).

[0100] The cell 800A includes four active regions 821-824 arranged consecutively along the Y-Y' axis, and has a height along the Y-Y' axis corresponding to 2h. Similar to the cell 200, the cell 800A includes, in one or more embodiments, no more than three rows of M0 conductive patterns over each cell height h. In the example configuration in FIG. 8A, the cell 800A includes four rows of M0 conductive patterns. In some embodiments, a maximum number of rows of M0 conductive patterns in the cell 800A is six. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell 800A.

[0101] In FIG. 8B, the cell 800B includes a different layout diagram for an AOI22D1 cell. A difference between the cell 800A and the cell 800B is the arrangement of PMOS and NMOS active regions along the Y-Y' axis. In FIG. 8A, the active regions are arranged in the following order PMOS, PMOS, NMOS, NMOS, whereas FIG. 8B, the active regions are arranged in the following order PMOS, NMOS, PMOS, NMOS. A further difference involves a structure for realizing the node C in FIG. 7.

[0102] The cell 800B comprises active regions 871-874 extending along the X-X' axis. The active regions 871 is over an N well 876, and the active region 873 is over an N well 877. The cell 800B comprises functional gate regions 830, 831, 832, 833 over the active regions 871-874. The gate regions 830, 832 are aligned along the Y-Y' axis and are electrically isolated from each other. The gate regions 831, 833 are aligned along the Y-Y' axis and are electrically isolated from each other. The gate regions 830, 831 configure, together with the active region 871, transistors PB2, PB1 as described with respect to FIG. 7. The functional gate regions 830, 831 configure, together with the active region 872, transistors NB2, NB1 as described with respect to FIG. 7. The gate regions 832, 833 configure, together with the active region 873, transistors PA2, PA1 as described with respect to FIG. 7. The functional gate regions 832, 833 configure, together with the active region 874, transistors NA2, NA1 as described with respect to FIG. 7. The PMOS and NMOS transistors of the cell 800B are electrically coupled together by various MD contact structures, VG vias, VD vias, M0 conductive patterns, V0 vias and M1 conductive patterns to realize the circuitry described with respect to FIG. 7. The PMOS and NMOS transistors of the cell 800B are further electrically coupled to back side VDD and VSS power rails by various VB through vias to realize the circuitry described with respect to FIG. 7.

[0103] To realize node C in FIG. 7, the cell 800B comprises, instead of the elongated MD contact

structure MD81 as in FIG. 8A, an extended contact structure 879 which electrically couples the corresponding source/drain regions of the transistors PB2, PB1, PA2, PA1. The extended contact structure 879 is schematically illustrated in the drawings with the label “Fly-MD,” and is referred to herein as “Fly-MD structure.” Opposite ends of the Fly-MD structure 879 are electrically coupled to corresponding MD contact structures in the active regions 871, 873, by corresponding VD2 vias 880, 881. A middle portion of the Fly-MD structure 879 extends over, or “flies” over, a underlying MD contact structure in the active region 872, without being electrically coupled to the underlying MD contact structure in the active region 872. In some embodiments, Fly-MD structures belong to a mask different from a mask for MD contact structures. An example structure for the Fly-MD structure 879 is described with respect to FIG. 9.

[0104] The cell 800B comprises M1 conductive patterns M181-M184, M186 corresponding to the inputs A1, A2, B1, B2 and the output ZN described with respect to FIG. 7. The M1 conductive patterns M181-M184, M186 are IO patterns of the cell 800B and are arranged in a boundary 860 having the same shape as described with respect to FIG. 8A. Besides the boundary 860, which contains the IO patterns and corresponds to the second boundary 250, the cell 800B further comprises a further boundary corresponding to the first boundary 220. The further boundary, also referred to as the first boundary of the cell 800B, is rectangular, and has the same shape as described with respect to FIG. 8A. For simplicity, the first, rectangular boundary of the cell 800B is omitted in FIG. 8B. The first boundary contains therein various layers and features of the cell 800B on the front side of a substrate up to, but not including, the pin access layer (i.e., the M1 layer).

[0105] The cell 800B includes four active regions 871-874 arranged consecutively along the Y-Y' axis, and has a height along the Y-Y' axis corresponding to 2h. Similar to the cell 200, the cell 800B includes, in one or more embodiments, no more than three rows of M0 conductive patterns over each cell height h. In the example configuration in FIG. 8B, the cell 800B includes four rows of M0 conductive patterns. In some embodiments, a maximum number of rows of M0 conductive patterns in the cell 800B is six. In at least one embodiment, one or more advantages described herein are achievable in an IC device corresponding to the cell 800B.

[0106] FIG. 9 is a schematic cross-sectional view, taken along lines IX-IX in FIG. 8B, of an IC device 900, in accordance with some embodiments. The IC device 900 comprises, in FIG. 9, an AOI22D1 logic corresponding to the cell 800B described with respect to FIG. 8B. Components in FIG. 9 having corresponding components in FIG. 8B are designated by the reference numerals of FIG. 8B increased by 100. Components in FIG. 9 having corresponding components in FIG. 3 are designated by the reference numerals of FIG. 3 increased by 600.

[0107] As shown in FIG. 9, the IC device 900 comprises a substrate 960 over which circuit elements and structures corresponding to the cell 800B are formed. The IC device 900 comprises, under the substrate 960, BM conductive patterns 934, 935, 936 configured to, e.g., provide power supply voltages to circuit elements on the front side of the substrate 960. For example, the BM conductive pattern 934 is a VDD power rail, and the BM conductive pattern 935 is a VSS power rail. The IC device 900 comprises, over the substrate 960, N wells 976, 977 corresponding to the N wells 876, 877 in FIG. 8B. The IC device 900 further comprises P well 971, N well 972, P well 973 corresponding to source/drain regions in the active regions 871, 872, 873 in FIG. 8B. MD contact structures 991, 992, 993 are over and in electrical contact with the corresponding source/drain regions 971, 972, 973. The IC device 900 comprises a Fly-MD structure 979 corresponding to the Fly-MD structure 879 in FIG. 8B. Opposite ends of the Fly-MD structure 979 are electrically coupled to the corresponding MD contact structures 991, 993 over the active regions 971, 973, by corresponding VD2 vias 980, 981. The VD2 vias 980, 981 correspond to the VD2 vias 880, 881 in FIG. 8B. A middle portion of the Fly-MD structure 979 extends over, or “flies” over, without being electrically coupled to, the MD contact structure 992 over the active region 972. The IC device 900 further comprises an interconnect structure 968 with various metal layers and via layers. Example M0 conductive patterns 994, 995 in the interconnect structure 968 are illustrated in FIG. 9. In at

least one embodiment, one or more advantages described herein are achievable in the IC device **900**.

[0108] FIG. **10A** is a schematic cross-sectional view of an IC layout diagram of an IC device **1000A** including layout diagrams of various cells constituting the IC layout diagram of the IC device **1000A**, in accordance with some embodiments.

[0109] The IC device **1000A** comprises cells **1001-1008**. The cells **1001, 1004, 1005, 1008** are ND2D1 cells corresponding to the cell **500A** described with respect to FIG. **5A**. The cells **1002, 1007** are INV4 cells corresponding to the cell **600A** described with respect to FIG. **6A**. The cells **1003, 1006** are INV1 cells corresponding to the cell **200** described with respect to FIGS. **2A-2D**. For simplicity, the first boundaries containing layers and features under the pin access layer of the cells are omitted, and the second boundaries containing IO patterns in the pin access layer of the cells are shown.

[0110] Along the Y-Y' axis, one of the cells **1001-1008** is placed in abutment with another of the cells **1001-1008**, at a common edge extending along the X-X' axis. For example, the cell **1001** is placed in abutment with the cell **1005** at a common edge **1011** extending along the X-X' axis. In this abutment, the first, rectangular boundaries of the cells **1001, 1005** abut each other at the common edge **1011**. At the same time, the second, hexagonal boundaries of the cells **1001, 1005** also abut each other at the common edge **1011**.

[0111] Along the X-X' axis, one of the cells **1001-1008** is placed in abutment with another of the cells **1001-1008**, at a corner of the corresponding second boundaries of the cells. For example, the second, hexagonal boundaries of the cells **1001, 1002** abut each other at a corner **1012**. The first rectangular boundaries of the cells **1001, 1002** are spaced from each other. For example, an edge **1013** of the first rectangular boundary of the cell **1001** is separated along the X-X' axis from an adjacent edge **1014** of the first rectangular boundary of the cell **1002** by an empty space. In at least one embodiment, this empty space is unfilled. In one or more embodiments, this empty space is filled by an appropriate filler cell as described herein. In at least one embodiment, one or more advantages described herein are achievable in the IC device **1000A**.

[0112] FIG. **10B** is a schematic cross-sectional view of an IC layout diagram of an IC device **1000B** including layout diagrams of various cells constituting the IC layout diagram of the IC device **1000B**, in accordance with some embodiments. A difference between the IC device **1000A** and the IC device **1000B** is the arrangement of PMOS and NMOS active regions along the Y-Y' axis. In FIG. **10A**, the active regions are arranged in the following order PMOS, NMOS, PMOS, NMOS, whereas FIG. **10B**, the active regions are arranged in the following order PMOS, PMOS, NMOS, NMOS.

[0113] The IC device **1000B** comprises the cells **1001-1004** as in the IC device **1000A**. The IC device **1000B** further comprises cells **1025-1028**. The cells **1025, 1028** are ND2D1 cells corresponding to the cell **500A** described with respect to FIG. **5A**, but with the PMOS and NMOS active regions switch place. The cell **1026** is an INV1 corresponding to the cell **200** described with respect to FIGS. **2A-2D**, but with the PMOS and NMOS active regions switch place. The cell **1027** is an INV4 cell corresponding to the cell **600A** described with respect to FIG. **6A**, but with the PMOS and NMOS active regions switch place. The cells in the IC device **1000B** are placed in abutment in the same manner as described with respect to FIG. **10A**. In at least one embodiment, one or more advantages described herein are achievable in the IC device **1000B**.

[0114] FIG. **10C** is a schematic cross-sectional view of an IC layout diagram of an IC device **1000C** including layout diagrams of various cells constituting the IC layout diagram of the IC device **1000C**, in accordance with some embodiments. A difference between the IC device **1000C** and the IC devices **1000A, 1000B** is that cell abutment along the X-X' axis in the IC device **1000C** occur at common edges, rather than at corners.

[0115] The IC device **1000C** comprises cells **1041-1044**. The cells **1041, 1042** are ND2D1 cells corresponding to the cell **500A** described with respect to FIG. **5A**. The cell **1043** is an INV4 cell

corresponding to the cell **600A** described with respect to FIG. 6A, but with the PMOS and NMOS active regions switch place. The cells **1044** is an INV4 cell with a double cell height and corresponding to the cell **600B** described with respect to FIG. 6B. For simplicity, the first boundaries containing layers and features under the pin access layer of the cells are omitted, and the second boundaries containing IO patterns in the pin access layer of the cells are shown.

[0116] Along the Y-Y' axis, one of the cells **1041-1044** is placed in abutment with another of the cells **1041-1044**, at a common edge extending along the X-X' axis. For example, the cell **1041** is placed in abutment with the cell **1042** at a common edge **1051** extending along the X-X' axis, as described with respect to FIG. 10A.

[0117] Along the X-X' axis, one of the cells **1041-1044** is placed in abutment with another of the cells **1041-1044**, at one or more common edges for both the first boundary and the second boundary. For example, the second, hexagonal boundaries of the cells **1041**, **1043** abut each other at a common edge **1052** oblique to both the X-X' axis and the Y-Y' axis. At the same time, the first rectangular boundaries of the cells **1041**, **1042** abut each other at a common edge **1053**. In at least one embodiment, one or more advantages described herein are achievable in the IC device **1000C**.

[0118] FIG. 10D is a schematic cross-sectional view of an IC layout diagram of an IC device **1000D** including layout diagrams of various cells constituting the IC layout diagram of the IC device **1000D**, in accordance with some embodiments. In the IC device **1000D**, cell abutments along the X-X' axis are performed in a mixed approach, using both cell abutments at common edges as in FIG. 10C, and cell abutments at corners as in FIGS. 10A-10B.

[0119] The IC device **1000D** comprises cells **1061-1069**. The cells **1061**, **1064**, **1066** are ND2D1 cells corresponding to the cell **500A** described with respect to FIG. 5A. The cell **1062** is an INV4 cell corresponding to the cell **600A** described with respect to FIG. 6A, but with the PMOS and NMOS active regions switch place. The cells **1063**, **1065**, **1069** are INV1 cells corresponding to the cell **200** described with respect to FIGS. 2A-2D. The cell **1067** is an INV4 cell corresponding to the cell **600A** described with respect to FIG. 6A, but with the PMOS and NMOS active regions switch place. The cell **1068** is an ND2D1 cell corresponding to the cell **500A** described with respect to FIG. 5A, but with the PMOS and NMOS active regions switch place. In the IC device **1000D**, cell abutments along the X-X' axis are performed in a mixed approach, using both cell abutments at common edges and cell abutments at corners. For example, the cells **1061**, **1062** abut each other at common edges as described with respect to the cells **1041**, **1043** in FIG. 10C. For another example, the cells **1063**, **1064** abut each other at a corner as described with respect to the cells **1001**, **1002** in FIG. 10A. In at least one embodiment, one or more advantages described herein are achievable in the IC device **1000D**.

[0120] FIG. 10E is a schematic cross-sectional view of an IC layout diagram of an IC device **1000E** including layout diagrams of various cells constituting the IC layout diagram of the IC device **1000E**, in accordance with some embodiments. The IC device **1000E** is a further example in which cell abutments along the X-X' axis are performed in a mixed approach, using both cell abutments at common edges as in FIG. 10C, and cell abutments at corners as in FIGS. 10A-10B.

[0121] The IC device **1000E** comprises cells **1081-1088**. The cells **1081**, **1083**, **1084**, **1086** are ND2D1 cells corresponding to the cell **500A** described with respect to FIG. 5A. The cell **1082** is an INV4 cell corresponding to the cell **600A** described with respect to FIG. 6A, but with the PMOS and NMOS active regions switch place. The cells **1085**, **1087** are INV6 cells corresponding to the cell **600C** described with respect to FIG. 6C. The cell **1088** is an ND2D1 cell corresponding to the cell **500A** described with respect to FIG. 5A, but with the PMOS and NMOS active regions switch place. In the IC device **1000E**, cell abutments along the X-X' axis are performed in a mixed approach, using both cell abutments at common edges and cell abutments at corners. For example, the cells **1081**, **1082** abut each other at common edges as described with respect to the cells **1041**, **1043** in FIG. 10C. For another example, the cells **1083**, **1084** abut each other at a corner as described with respect to the cells **1001**, **1002** in FIG. 10A. In at least one embodiment, one or

more advantages described herein are achievable in the IC device **1000E**.

[0122] FIGS. **11A-11I** are schematic cross-sectional views of layout diagrams of various filler cells, in accordance with some embodiments. As described herein and illustrated in FIGS. **10A-10E**, cell abutments in a layout diagram of an IC device create empty spaces which are filled by one or more appropriate filler cells.

[0123] In FIG. **11A**, a filler cell **FILL 1** comprises a portion of a PMOS active region **1101** over an N well **1102**, gate regions **1103**, **1104**, an MD contact structure **1105**, a BM conductive pattern **1106**, and a boundary **1107**. The boundary **1107** has a shape of a triangle with edges oblique to both the X-X' axis and Y-Y' axis. The gate regions **1103**, **1104** are dummy gate regions in one or more embodiments. In at least one embodiment, the filler cell **FILL 1** comprises an NMOS active region instead of the PMOS active region **1101**.

[0124] In FIG. **11B**, a filler cell **FILL 2** comprises a portion of a PMOS active region **1111** over an N well **1112**, gate regions **1113**, **1114**, **1118**, MD contact structures **1115**, a BM conductive pattern **1116**, and a boundary **1117**. The boundary **1117** has a shape of an isosceles trapezoid with side edges oblique to both the X-X' axis and Y-Y' axis. The gate regions **1113**, **1114** are dummy gate regions in one or more embodiments. In at least one embodiment, the filler cell **FILL 2** comprises an NMOS active region instead of the PMOS active region **1111**. In at least one embodiment, the filler cell **FILL 2** is configured as a decap cell, for example, with the gate region **1118** being a functional gate region.

[0125] In FIG. **11C**, a filler cell **FILL 3** comprises a portion of a PMOS active region **1121** over an N well **1122**, gate regions **1123**, **1124**, **1128**, **1129**, MD contact structures **1125**, a BM conductive pattern **1126**, and a boundary **1127**. The boundary **1127** has a shape of an isosceles trapezoid with side edges oblique to both the X-X' axis and Y-Y' axis. The gate regions **1123**, **1124** are dummy gate regions in one or more embodiments. In at least one embodiment, the filler cell **FILL 3** comprises an NMOS active region instead of the PMOS active region **1121**. In at least one embodiment, the filler cell **FILL 3** is configured as a decap cell, for example, with the gate regions **1128**, **1129** being functional gate regions. Other filler cells having similar shapes to the filler cell **FILL 3** and more than four gate regions are within the scopes of various embodiments.

[0126] In FIG. **11D**, a filler cell **FILL 4** comprises a portion of an NMOS active region **1131**, gate regions **1133**, **1134**, an MD contact structure **1135**, a BM conductive pattern **1136**, and a boundary **1137**. The boundary **1137** has a shape of a triangle with edges oblique to both the X-X' axis and Y-Y' axis. The gate regions **1133**, **1134** are dummy gate regions in one or more embodiments. In at least one embodiment, the filler cell **FILL 4** comprises a PMOS active region instead of the NMOS active region **1131**.

[0127] In FIG. **11E**, a filler cell **FILL 5** comprises a portion of an NMOS active region **1141**, gate regions **1143**, **1144**, **1148**, MD contact structures **1145**, a BM conductive pattern **1146**, and a boundary **1147**. The boundary **1147** has a shape of an isosceles trapezoid with side edges oblique to both the X-X' axis and Y-Y' axis. The gate regions **1143**, **1144** are dummy gate regions in one or more embodiments. In at least one embodiment, the filler cell **FILL 5** comprises a PMOS active region instead of the NMOS active region **1141**. In at least one embodiment, the filler cell **FILL 5** is configured as a decap cell, for example, with the gate region **1148** being a functional gate region.

[0128] In FIG. **11F**, a filler cell **FILL 6** comprises a portion of an NMOS active region **1151**, gate regions **1153**, **1154**, **1158**, **1159**, MD contact structures **1155**, a BM conductive pattern **1156**, and a boundary **1157**. The boundary **1157** has a shape of an isosceles trapezoid with side edges oblique to both the X-X' axis and Y-Y' axis. The gate regions **1153**, **1154** are dummy gate regions in one or more embodiments. In at least one embodiment, the filler cell **FILL 6** comprises a PMOS active region instead of the NMOS active region **1151**. In at least one embodiment, the filler cell **FILL 6** is configured as a decap cell, for example, with the gate regions **1158**, **1159** being functional gate regions. Other filler cells having similar shapes to the filler cell **FILL 6** and more than four gate regions are within the scopes of various embodiments.

[0129] Each of the filler cells FILL 1 to FILL 6 has one active region along the Y-Y' axis, and is considered to have a half cell height ($h/2$). Filler cells with one cell height (h) in accordance with some embodiments are described herein.

[0130] In FIG. 11G, a filler cell FILL 7 is obtained by combining a filler cell FILL 1 with a filler cell FILL 4. The filler cell FILL 7 has one cell height (h) and a boundary 1167 which has a shape of a rhombus. In at least one embodiment, the PMOS active region and the NMOS active region in the filler cell FILL 7 switch place.

[0131] In FIG. 11H, a filler cell FILL 8 is obtained by combining a filler cell FILL 2 with a filler cell FILL 5. The filler cell FILL 8 has one cell height (h) and a boundary 1177 which has a shape of a hexagon. In at least one embodiment, the PMOS active region and the NMOS active region in the filler cell FILL 8 switch place.

[0132] In FIG. 11I, a filler cell FILL 9 is obtained by combining a filler cell FILL 3 with a filler cell FILL 6. The filler cell FILL 9 has one cell height (h) and a boundary 1187 which has a shape of a hexagon. In at least one embodiment, the PMOS active region and the NMOS active region in the filler cell FILL 9 switch place. In at least one embodiment, the filler cell FILL 8 or FILL 9 is configured as a decap cell. Other filler cells having similar shapes to the filler cell FILL 9 and more than four columns of gate regions are within the scopes of various embodiments.

[0133] FIG. 12 is a schematic cross-sectional view of an IC layout diagram of an IC device 1200 including layout diagrams of various cells constituting the IC layout diagram of the IC device 1200, in accordance with some embodiments.

[0134] The IC device 1200 comprises the cells 1081-1088 of the IC device 1000E. The IC device 1200 further comprises filler cells 1201-1206, 1211-1212, 1221-1224, 1231-1235, 1241-1248 that fill in the empty spaces between the cells 1081-1088. Each of the filler cells 1201-1206, 1211-1212, 1221-1224, 1231-1235, 1241-1248 is one of the filler cells FILL 1 to FILL 9 described with respect to FIGS. 11A-11I. For example, filler cell 1201 corresponds to filler cell FILL 3. Filler cell 1202 corresponds to filler cell FILL 4 with a PMOS active region. Filler cells 1203-1206 correspond to filler cell FILL 1. Filler cell 1211 corresponds to filler cell FILL 7. Filler cell 1212 corresponds to filler cell FILL 6 with a PMOS active region. Filler cell 1221 corresponds to filler cell FILL 6. Filler cells 1222-1223 correspond to filler cell FILL 1 with an NMOS active region. Filler cell 1224 corresponds to filler cell FILL 4. Filler cells 1231, 1233 correspond to filler cell FILL 6 with a PMOS active region. Filler cells 1232, 1235 correspond to filler cell FILL 1. Filler cell 1234 corresponds to filler cell FILL 4 with a PMOS active region. Filler cells 1241, 1243 correspond to filler cell FILL 3 with an NMOS active region. Filler cells 1242, 1244, 1245 correspond to filler cell FILL 4. Filler cell 1246 corresponds to filler cell FILL 2 with an NMOS active region. Filler cell 1247 corresponds to filler cell FILL 5. Filler cell 1248 corresponds to filler cell FILL 9 with the PMOS active region and the NMOS active region switching place. Each of the filler cells 1201-1206, 1211-1212, 1221-1224, 1231-1235, 1241-1248 are placed in abutment with one or more of the cells 1081-1088, or with another filler cell by cell abutment at common edges as described with respect to FIG. 10C. In at least one embodiment, one or more advantages described herein are achievable in the IC device 1200.

[0135] FIG. 13A is a flowchart of a method 1300A of generating a layout diagram and using the layout diagram to manufacture an IC device, in accordance with some embodiments.

[0136] Method 1300A is implementable, for example, using an EDA system as described herein and/or an integrated circuit (IC) manufacturing system as described herein, in accordance with some embodiments. Regarding method 1300A, examples of the layout diagram include the layout diagrams disclosed herein with respect to one or more of FIGS. 1-12, or the like. Examples of an IC device to be manufactured according to method 1300A include the IC devices disclosed with respect to one or more of FIGS. 1-12.

[0137] At block 1305, a layout diagram is generated which, among other things, include patterns represent one or more cells as described with respect to one or more of FIGS. 1-12, or the like.

Block **1305** is discussed in more detail below with respect to FIG. **13B**.

[0138] At block **1315**, based on the layout diagram, at least one of (A) one or more photolithographic exposures are made or (b) one or more semiconductor masks are fabricated or (C) one or more components in a layer of an IC device are fabricated. Block **1315** is discussed in more detail below with respect to FIG. **13C**.

[0139] FIG. **13B** is a flowchart of a method **1300B** of generating a layout diagram, in accordance with some embodiments. More particularly, the flowchart of FIG. **13B** shows additional blocks that demonstrates one example of procedures implementable in block **1305** of FIG. **13A**, in accordance with one or more embodiments. In FIG. **13B**, block **1305** includes blocks **1325**, **1335**.

[0140] At block **1325**, at least one cell is generated, or retrieved from a cell library. The at least one generated or retrieved cell has at least one gate region across at least one active region in a first boundary, and, in a second boundary, at least one input/output (IO) pattern configured to electrically couple one or more of the at least one active region and the at least one gate region to external circuitry outside the first cell. A first edge of the first boundary and a second edge of the second boundary overlap each other and form therebetween an acute angle. For example, at least one cell corresponding to one or more of the cells **200**, **500A-500C**, **600A-600C**, **800A-800B** is generated, or retrieved from a cell library. The generated or retrieved cell, in the example configuration in FIG. **2C**, has at least one gate region **210** across at least one active region **201**, **202** in a first boundary **220**. In a second boundary **250**, the cell has at least one input/output (IO) pattern **245**, **246** configured to electrically couple one or more of the at least one active region **210** and the at least one gate region **201**, **202** to external circuitry outside the cell. A first edge **222** of the first boundary **220** and a second edge **253** of the second boundary **250** overlap each other and form therebetween an acute angle **257**.

[0141] In some embodiments, the at least one generated or retrieved cell has at least one gate region across at least one active region, a conductive pattern overlapping and configured to be electrically coupled to the at least one active region or the at least one gate region, and an IO pattern overlapping and configured to be electrically coupled to the conductive pattern. The IO pattern and the at least one gate region form an acute angle therebetween. For example, at least one cell corresponding to one or more of the cells **200**, **500A-500C**, **600A-600C**, **800A-800B** is generated, or retrieved from a cell library. The generated or retrieved cell, in the example configuration in FIG. **2D**, has at least one gate region **210** across at least one active region **201**, **202**, an M0 conductive pattern **241**, **242** overlapping and configured to be electrically coupled to the at least one active region **210** or the at least one gate region **201**, **202**, and an IO pattern **245**, **246** overlapping and configured to be electrically coupled to the M0 conductive pattern **241**, **242**. The IO pattern **245**, **246** and the at least one gate region **210** form an acute angle **257** therebetween.

[0142] At block **1335**, the at least one generated or retrieved cell is placed in abutment with another cell in the layout diagram. In some embodiments, the abutment is at a corner of corresponding boundaries of the cells as described with respect to FIGS. **10A-10B**, or at a common edge of the corresponding boundaries of the cells as described with respect to FIG. **10C**, or in a mixed approach as described with respect to FIGS. **10D-10E**. In some embodiments, the other cell is a non-filler cell or a filler cell as described with respect to FIGS. **11A-11I** and **12**. In at least one embodiment, the generated layout diagram of the IC device is stored on a non-transitory computer-readable recording medium.

[0143] FIG. **13C** is a flowchart of a method **1300C** of fabricating one or more components of an IC device, based on the layout diagram, in accordance with some embodiments. More particularly, the flowchart of FIG. **13C** shows additional blocks that demonstrates one example of procedures implementable in block **1315** of FIG. **13A**, in accordance with one or more embodiments. In FIG. **13C**, block **1315** includes blocks **1345**, **1355**, **1365**.

[0144] At block **1345**, at least one gate region across at least one active region over a substrate. An example manufacturing process starts from a substrate, such as the substrate **360** described with

respect to FIG. 3. The substrate comprises, in at least one embodiment, silicon, silicon germanium (SiGe), gallium arsenic, or other suitable semiconductor or dielectric materials. Active regions are formed in or over the substrate, using one or more masks corresponding to one or more active regions in the layout diagrams described herein. A gate dielectric material layer is deposited over the substrate. Example materials of the gate dielectric material layer include, but are not limited to, a high-k dielectric layer, an interfacial layer, and/or combinations thereof. In some embodiments, the gate dielectric material layer is deposited over the substrate by atomic layer deposition (ALD) or other suitable techniques. A gate electrode layer is deposited over the gate dielectric material layer. Example materials of the gate electrode layer include, but are not limited to, polysilicon, metal, Al, AlTi, Ti, TiN, TaN, Ta, TaC, TaSiN, W, WN, MON, and/or other suitable conductive materials. In some embodiments, the gate electrode layer is deposited by chemical vapor deposition (CVD), physical vapor deposition (PVD or sputtering), plating, atomic layer deposition (ALD), and/or other suitable processes. A patterning process is then performed, using one or more masks corresponding to one or more gate electrodes in the layout diagrams described herein. As a result, the gate dielectric material layer is patterned in to one or more gate dielectric layers, such as the gate dielectric layers **363**, **364**, and the gate electrode layer is patterned into one or more gate electrodes, such as the gate electrode **310** or the dummy gate regions **318**, **319** described with respect to FIG. 3. In at least one embodiment, spacers are formed, by deposition and patterning, on opposite sides of each gate electrode. Example materials of the spacers include, but are not limited to, silicon nitride, oxynitride, silicon carbide and other suitable materials. Example deposition processes include, but are not limited to, plasma enhanced chemical vapor deposition (PECVD), low-pressure chemical vapor deposition (LPCVD), sub-atmospheric chemical vapor deposition (SACVD), atomic layer deposition (ALD), or the like. Example patterning processes include, but are not limited to, a wet etch process, a dry etch process, or combinations thereof. Drain/source regions, such as the drain/source regions **304**, **306** described with respect to FIG. 3, are formed in the active regions of the substrate. In at least one embodiment, the drain/source regions are formed by using the gate electrodes and the spacers as a mask. For example, the formation of the drain/source regions is performed by an ion implantation or a diffusion process. Depending on the type of the devices or transistors, the drain/source regions are doped with p-type dopants, such as boron or BF₂, n-type dopants, such as phosphorus or arsenic, and/or combinations thereof. A conductive layer, e.g., a metal, is deposited over the substrate, thereby making electrical connections to the drain/source regions. A planarizing process is performed to planarize the conductive layer, resulting in contact structures, such as the contact structures **336**, **337** described with respect to FIG. 3, in electrical contact with the underlying drain/source regions. The planarizing process comprises, for example, a chemical mechanical polish (CMP) process. A dielectric layer is deposited over the substrate with the drain/source contacts formed thereon. The dielectric layer is etched, and the etched portions are filled with a conductive material, such as a metal, to form one or more VD and VG vias, such as the VG via **338** described with respect to FIG. 3. A planarizing process is performed to obtain a planarized structure.

[0145] At block **1355**, deposition and patterning are performed to form, in a metal layer over the at least one gate region, a conductive pattern overlapping and electrically coupled to the at least one active region or the at least one gate region. In an example, an M0 layer including a conductive material, such as a metal, is deposited over the planarized structure and patterned to form various M0 conductive patterns, such as the M0 conductive pattern **342** described with respect to FIG. 3. In some embodiments, deposition and etching are performed to form at least one V0 via over and in electrical contact with the at least one M0 conductive patterns. In an example process, a dielectric layer is deposited over the patterned M0 layer. The dielectric layer is etched, and the etched portions are filled with a conductive material, such as a metal, to form one or more vias in a V0 layer. For example, the V0 layer comprises the V0 via **344** described with respect to FIG. 3. A planarizing process is then performed.

[0146] At block **1365**, deposition and patterning are performed to form, in a pin access layer over the metal layer, an input/output (IO) pattern overlapping and electrically coupled to the conductive pattern. The IO pattern and the at least one gate region form an acute angle therebetween. For example, the pin access layer, e.g., the M1 layer, is deposited and patterned to form at least one M1 conductive pattern which extends obliquely to the gate regions, for example, as described with respect to FIGS. **2C-2D**, **5A-5C**, **6A-6C**, **8A-8B**. In an example process, an M1 layer including a conductive material, such as a metal, is deposited over the planarized structure obtained at the end of the formation of the V0 layer. The M1 layer is patterned to form at least one M1 conductive pattern, such as the M1 conductive pattern **346** described with respect to FIG. **3**. The M1 conductive pattern **346** is an IO pattern that is electrically coupled to the corresponding M0 conductive pattern **342** through the corresponding V0 via **344**, as described with respect to FIG. **3**.

[0147] In some embodiments, the process further comprises etching and depositing at least one through via structure and depositing and patterning a BM0 layer. In some embodiments, as described with respect to FIG. **3**, at least one through via is etched from a back side **362** of the substrate **360** toward a front side **361** of the substrate **360** which has transistors, e.g., **3NM**, thereon. A conductive material, e.g., a metal, is deposited in the etched via to form the VB through via **332**. The VB through via **332** is in electrical contact with a source/drain region **304** of at least one transistor, e.g., **3NM**. A BM0 layer is deposited on the back side **362** of the substrate **360**, and is patterned to form a BM0 conductive pattern **334** in electrical contact with the VB through via **332**. In at least one embodiment, the BM0 conductive pattern **334** is configured as a power rail.

[0148] The described methods include example operations, but they are not necessarily required to be performed in the order shown. Operations may be added, replaced, changed order, and/or eliminated as appropriate, in accordance with the spirit and scope of embodiments of the disclosure. Embodiments that combine different features and/or different embodiments are within the scope of the disclosure and will be apparent to those of ordinary skill in the art after reviewing this disclosure.

[0149] In some embodiments, some or all of the methods discussed above are performed by an IC layout diagram generation system. In some embodiments, an IC layout diagram generation system is usable as part of a design house of an IC manufacturing system discussed below.

[0150] FIG. **14** is a block diagram of an electronic design automation (EDA) system **1400** in accordance with some embodiments.

[0151] In some embodiments, EDA system **1400** includes an APR system. Methods described herein of designing layout diagrams represent wire routing arrangements, in accordance with one or more embodiments, are implementable, for example, using EDA system **1400**, in accordance with some embodiments.

[0152] In some embodiments, EDA system **1400** is a general purpose computing device including a hardware processor **1402** and a non-transitory, computer-readable storage medium **1404**. Storage medium **1404**, amongst other things, is encoded with, i.e., stores, computer program code **1406**, i.e., a set of executable instructions. Execution of instructions **1406** by hardware processor **1402** represents (at least in part) an EDA tool which implements a portion or all of the methods described herein in accordance with one or more embodiments (hereinafter, the noted processes and/or methods).

[0153] Processor **1402** is electrically coupled to computer-readable storage medium **1404** via a bus **1408**. Processor **1402** is also electrically coupled to an I/O interface **1410** by bus **1408**. A network interface **1412** is also electrically connected to processor **1402** via bus **1408**. Network interface **1412** is connected to a network **1414**, so that processor **1402** and computer-readable storage medium **1404** are capable of connecting to external elements via network **1414**. Processor **1402** is configured to execute computer program code **1406** encoded in computer-readable storage medium **1404** in order to cause system **1400** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **1402** is a central processing unit

(CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

[0154] In one or more embodiments, computer-readable storage medium **1404** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **1404** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **1404** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

[0155] In one or more embodiments, storage medium **1404** stores computer program code **1406** configured to cause system **1400** (where such execution represents (at least in part) the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **1404** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **1404** stores library **1407** of standard cells including such standard cells as disclosed herein.

[0156] EDA system **1400** includes I/O interface **1410**. I/O interface **1410** is coupled to external circuitry. In one or more embodiments, I/O interface **1410** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **1402**.

[0157] EDA system **1400** also includes network interface **1412** coupled to processor **1402**. Network interface **1412** allows system **1400** to communicate with network **1414**, to which one or more other computer systems are connected. Network interface **1412** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more systems **1400**.

[0158] System **1400** is configured to receive information through I/O interface **1410**. The information received through I/O interface **1410** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **1402**. The information is transferred to processor **1402** via bus **1408**. EDA system **1400** is configured to receive information related to a UI through I/O interface **1410**. The information is stored in computer-readable recording medium **1404** as user interface (UI) **1442**.

[0159] In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by EDA system **1400**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

[0160] In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

[0161] FIG. **15** is a block diagram of an integrated circuit (IC) manufacturing system **1500**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some

embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using manufacturing system **1500**.

[0162] In FIG. **15**, IC manufacturing system **1500** includes entities, such as a design house **1520**, a mask house **1530**, and an IC manufacturer/fabricator (“fab”) **1550**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1560**. The entities in system **1500** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **1520**, mask house **1530**, and IC fab **1550** is owned by a single larger company. In some embodiments, two or more of design house **1520**, mask house **1530**, and IC fab **1550** coexist in a common facility and use common resources.

[0163] Design house (or design team) **1520** generates an IC design layout diagram **1522**. IC design layout diagram **1522** includes various geometrical patterns designed for an IC device **1560**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1560** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **1522** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1520** implements a proper design procedure to form IC design layout diagram **1522**. The design procedure includes one or more of logic design, physical design or place-and-route operation. IC design layout diagram **1522** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **1522** can be expressed in a GDSII file format or DFII file format.

[0164] Mask house **1530** includes data preparation **1532** and mask fabrication **1544**. Mask house **1530** uses IC design layout diagram **1522** to manufacture one or more masks **1545** to be used for fabricating the various layers of IC device **1560** according to IC design layout diagram **1522**. Mask house **1530** performs mask data preparation **1532**, where IC design layout diagram **1522** is translated into a representative data file (“RDF”). Mask data preparation **1532** provides the RDF to mask fabrication **1544**. Mask fabrication **1544** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **1545** or a semiconductor wafer **1553**. The design layout diagram **1522** is manipulated by mask data preparation **1532** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1550**. In FIG. **15**, mask data preparation **1532** and mask fabrication **1544** are illustrated as separate elements. In some embodiments, mask data preparation **1532** and mask fabrication **1544** can be collectively referred to as mask data preparation.

[0165] In some embodiments, mask data preparation **1532** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **1522**. In some embodiments, mask data preparation **1532** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

[0166] In some embodiments, mask data preparation **1532** includes a mask rule checker (MRC) that checks the IC design layout diagram **1522** that has undergone processes in OPC with a set of

mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **1522** to compensate for limitations during mask fabrication **1544**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

[0167] In some embodiments, mask data preparation **1532** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1550** to fabricate IC device **1560**. LPC simulates this processing based on IC design layout diagram **1522** to create a simulated manufactured device, such as IC device **1560**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **1522**.

[0168] It should be understood that the above description of mask data preparation **1532** has been simplified for the purposes of clarity. In some embodiments, data preparation **1532** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **1522** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **1522** during data preparation **1532** may be executed in a variety of different orders.

[0169] After mask data preparation **1532** and during mask fabrication **1544**, a mask **1545** or a group of masks **1545** are fabricated based on the modified IC design layout diagram **1522**. In some embodiments, mask fabrication **1544** includes performing one or more lithographic exposures based on IC design layout diagram **1522**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1545** based on the modified IC design layout diagram **1522**. Mask **1545** can be formed in various technologies. In some embodiments, mask **1545** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **1545** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **1545** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **1545**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1544** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **1553**, in an etching process to form various etching regions in semiconductor wafer **1553**, and/or in other suitable processes.

[0170] IC fab **1550** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **1550** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

[0171] IC fab **1550** includes fabrication tools **1552** configured to execute various manufacturing operations on semiconductor wafer **1553** such that IC device **1560** is fabricated in accordance with

the mask(s), e.g., mask **1545**. In various embodiments, fabrication tools **1552** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

[0172] IC fab **1550** uses mask(s) **1545** fabricated by mask house **1530** to fabricate IC device **1560**. Thus, IC fab **1550** at least indirectly uses IC design layout diagram **1522** to fabricate IC device **1560**. In some embodiments, semiconductor wafer **1553** is fabricated by IC fab **1550** using mask(s) **1545** to form IC device **1560**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **1522**. Semiconductor wafer **1553** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **1553** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

[0173] Details regarding an integrated circuit (IC) manufacturing system (e.g., system **1500** of FIG. **15**), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20150278429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

[0174] In some embodiments, an integrated circuit (IC) device comprises a substrate, at least one active region over the substrate, at least one gate region extending across the at least one active region, and a metal layer. The at least one active region and the at least one gate region configure a plurality of transistors electrically coupled into a circuit having an input and an output. The metal layer comprises a first conductive pattern electrically coupled to the input of the circuit, and a second conductive pattern electrically coupled to the output of the circuit. The first conductive pattern and the second conductive pattern extend obliquely to the at least one active region or the at least one gate region.

[0175] In some embodiments, an integrated circuit (IC) device comprises a substrate, at least one active region over the substrate, at least one gate region extending across the at least one active region, a first metal layer over the at least one gate region, and a second metal layer over the first metal layer. The first metal layer comprises at least one first conductive pattern electrically coupled to the at least one active region or the at least one gate region. The second metal layer comprises at least one second conductive pattern electrically coupled to the at least one first metal conductive pattern. The at least one second conductive pattern extends obliquely to the at least one first conductive pattern.

[0176] In some embodiments, a non-transitory computer-readable recording medium stores thereon a layout of a cell of an integrated circuit (IC) device. The cell comprises at least one active region, at least one gate region extending across the at least one active region, a conductive pattern configured to be electrically coupled to the at least one gate region or the at least one active region, and a boundary. The boundary contains an entirety of the conductive pattern, and at least a portion of one or more of the at least one active region and the at least one gate region. At least one of a first edge of the boundary or the conductive pattern extends obliquely to the at least one active region or the at least one gate region.

[0177] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit (IC) device, comprising: a substrate; and at least one active region over the substrate; at least one gate region extending across the at least one active region, wherein the at least one active region and the at least one gate region configure a plurality of transistors electrically coupled into a circuit having an input and an output; and a metal layer comprising: a first conductive pattern electrically coupled to the input of the circuit; and a second conductive pattern electrically coupled to the output of the circuit, wherein the first conductive pattern and the second conductive pattern extend obliquely to the at least one active region or the at least one gate region.
2. The IC device of claim 1, further comprising: a metal-zero (M0) layer over the at least one gate region, wherein the metal layer is an upper metal layer over the M0 layer.
3. The IC device of claim 2, wherein the M0 layer comprises at least one M0 conductive pattern electrically coupled to the at least one active region or the at least one gate region, and the first conductive pattern or the second conductive pattern in the upper metal layer is electrically coupled to the at least one M0 conductive pattern.
4. The IC device of claim 2, wherein the upper metal layer is a metal-one (M1) layer.
5. The IC device of claim 2, wherein the at least one active region comprises first and second active regions, the first and second active regions extend along a first axis, and are directly adjacent each other along a second axis transverse to the first axis, and the M0 layer has no more than three rows of M0 conductive patterns over and between the first and second active regions, the rows of M0 conductive patterns extending along the first axis and directly adjacent each other along the second axis.
6. The IC device of claim 2, wherein the at least one active region comprises first and second active regions, the first and second active regions extend along a first axis, and are directly adjacent each other along a second axis transverse to the first axis, and the M0 layer has a first row of one or more M0 conductive patterns over the first active region, the first row extending along the first axis, and a second row of one or more M0 conductive patterns over the second active region, the second row extending along the first axis and directly adjacent the first row along the second axis.
7. The IC device of claim 2, wherein the at least one active region comprises first, second and third active regions, the second active region arranged between the first and third active regions, the IC device further comprises: first, second and third contact structures over and in electrical contact with corresponding source/drain regions correspondingly in the first, second and third active regions; and an extended contact structure over and electrically coupled to the first and third contact structures, the extended contact structure extending over the second contact structure without being electrically coupled to the second contact structure, and the M0 layer is over the first, second and third contact structures, and the extended contact structure.
8. The IC device of claim 1, further comprising: at least one dummy gate region, wherein at least one of the first conductive pattern or the second conductive pattern extends obliquely to and overlaps the at least one dummy gate region.
9. An integrated circuit (IC) device, comprising: a substrate; at least one active region over the substrate; at least one gate region extending across the at least one active region; a first metal layer over the at least one gate region, the first metal layer comprising at least one first conductive pattern electrically coupled to the at least one active region or the at least one gate region; and a second metal layer over the first metal layer, the second metal layer comprising at least one second conductive pattern electrically coupled to the at least one first conductive pattern, wherein the at least one second conductive pattern extends obliquely to the at least one first conductive pattern.

10. The IC device of claim 9, wherein the first metal layer is a metal-zero (M0) layer, and the second metal layer is a metal-one (M1) layer.

11. The IC device of claim 9, wherein the at least one active region comprises first and second active regions, the first and second active regions extend along a first axis, and are directly adjacent each other along a second axis transverse to the first axis, and the first metal layer has no more than three rows of conductive patterns over and between the first and second active regions, the rows of conductive patterns extending along the first axis and directly adjacent each other along the second axis.

12. The IC device of claim 9, wherein the at least one active region comprises first and second active regions, the first and second active regions extend along a first axis, and are directly adjacent each other along a second axis transverse to the first axis, and the first metal layer has a first row of one or more conductive patterns over the first active region, the first row extending along the first axis, and a second row of one or more conductive patterns over the second active region, the second row extending along the first axis and directly adjacent the first row along the second axis.

13. The IC device of claim 9, wherein the at least one active region comprises first, second and third active regions, the second active region arranged between the first and third active regions, the IC device further comprises: first, second and third contact structures over and in electrical contact with corresponding source/drain regions correspondingly in the first, second and third active regions; and an extended contact structure over and electrically coupled to the first and third contact structures, the extended contact structure extending over the second contact structure without being electrically coupled to the second contact structure, and the first metal layer is over the first, second and third contact structures, and the extended contact structure.

14. The IC device of claim 9, further comprising: at least one dummy gate region, wherein the at least one second conductive pattern extends obliquely to and overlaps the at least one dummy gate region.

15. A non-transitory computer-readable recording medium storing thereon a layout of a cell of an integrated circuit (IC) device, said cell comprising: at least one active region; at least one gate region extending across the at least one active region; a conductive pattern configured to be electrically coupled to the at least one gate region or the at least one active region; and a boundary which contains an entirety of the conductive pattern, and at least a portion of one or more of the at least one active region and the at least one gate region, wherein at least one of a first edge of the boundary or the conductive pattern extends obliquely to the at least one active region or the at least one gate region.

16. The non-transitory computer-readable recording medium of claim 15, wherein said cell further comprises a contact structure over the at least one active region, and the at least one of the first edge of the boundary or the conductive pattern overlaps and extends obliquely to the contact structure.

17. The non-transitory computer-readable recording medium of claim 15, wherein said cell further comprises a metal-zero (M0) layer over the at least one gate region, the at least one active region comprises first and second active regions, the first and second active regions extend along a first axis, and are directly adjacent each other along a second axis transverse to the first axis, and the M0 layer has no more than three rows of M0 conductive patterns over and between the first and second active regions, the rows of M0 conductive patterns extending along the first axis and directly adjacent each other along the second axis.

18. The non-transitory computer-readable recording medium of claim 15, wherein said cell further comprises a metal-zero (M0) layer over the at least one gate region, the at least one active region comprises first and second active regions, the first and second active regions extend along a first axis, and are directly adjacent each other along a second axis transverse to the first axis, and the M0 layer has a first row of one or more M0 conductive patterns over the first active region, the first row extending along the first axis, and a second row of one or more M0 conductive patterns over

the second active region, the second row extending along the first axis and directly adjacent the first row along the second axis.

19. The non-transitory computer-readable recording medium of claim 15, wherein said cell further comprises: a metal-zero (M0) layer over the at least one gate region, and an upper metal layer over the M0 layer and comprising the conductive pattern.

20. The non-transitory computer-readable recording medium of claim 15, wherein both the first edge of the boundary and the conductive pattern extend obliquely to the at least one active region or the at least one gate region.
